

BFS Traversal

Step 1: Start at E

Legend

Complete

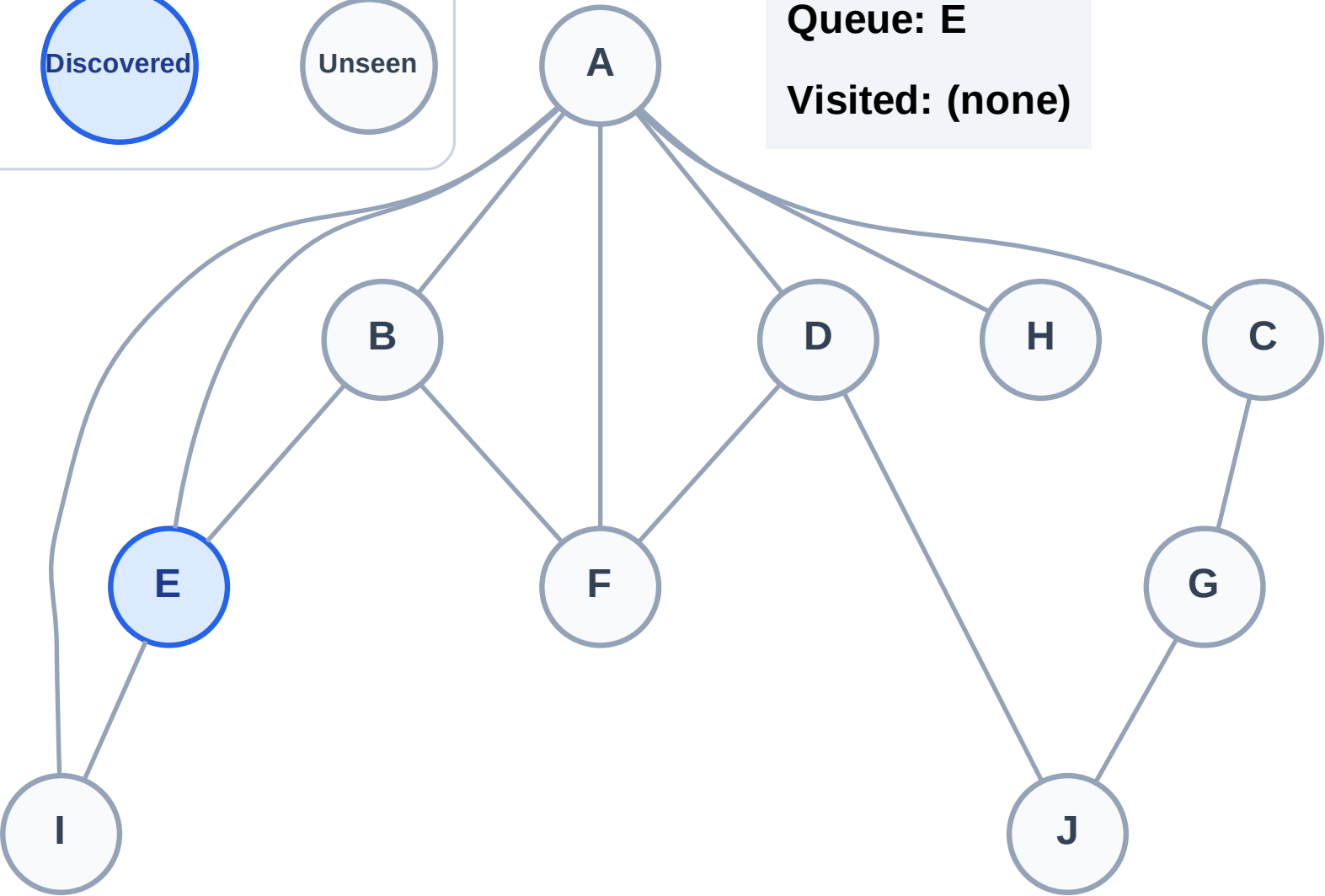
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

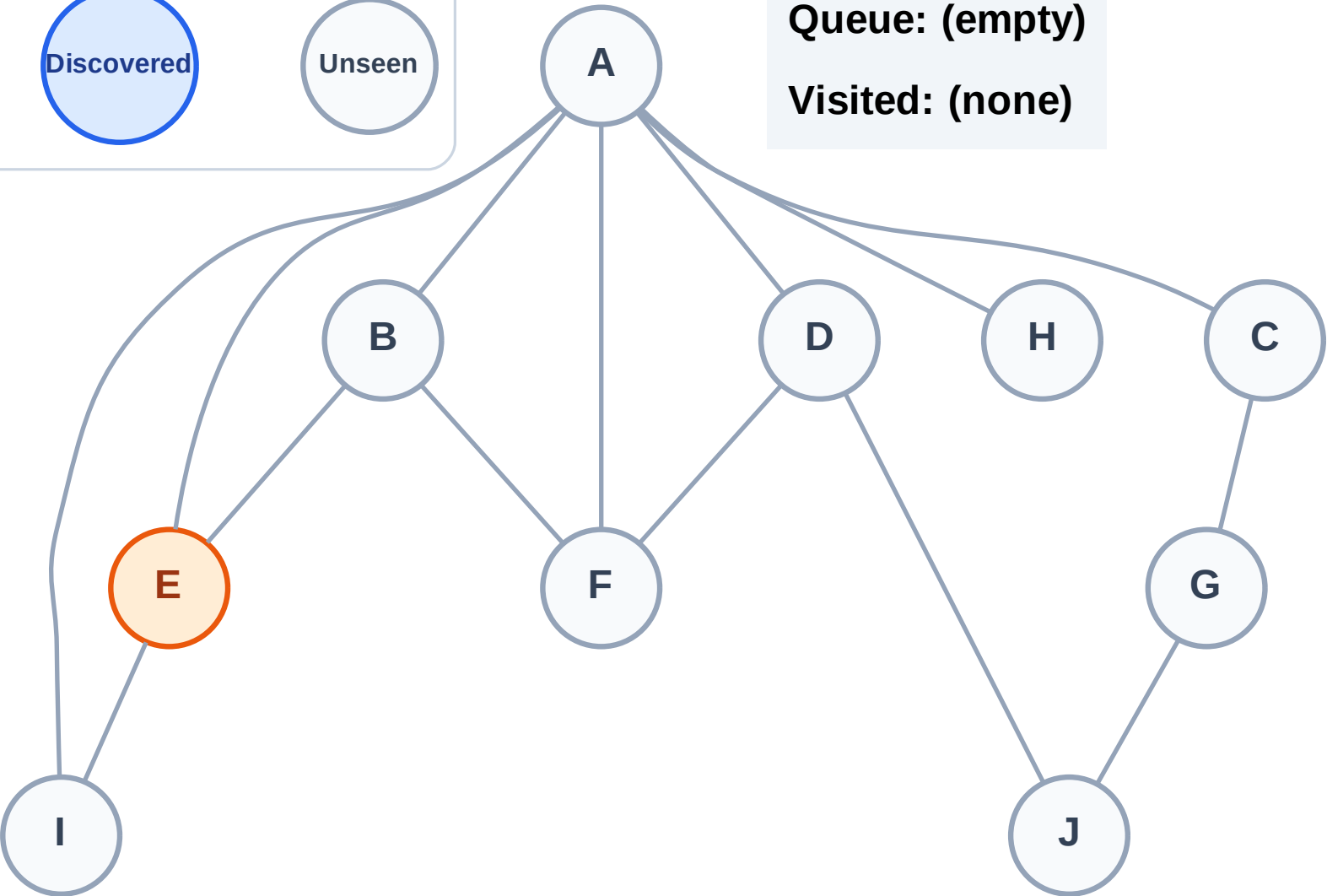
Complete

Processing

Discovered

Unseen

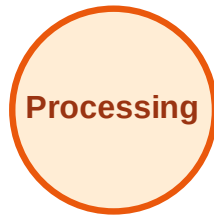
Queue: (empty)
Visited: (none)



BFS Traversal

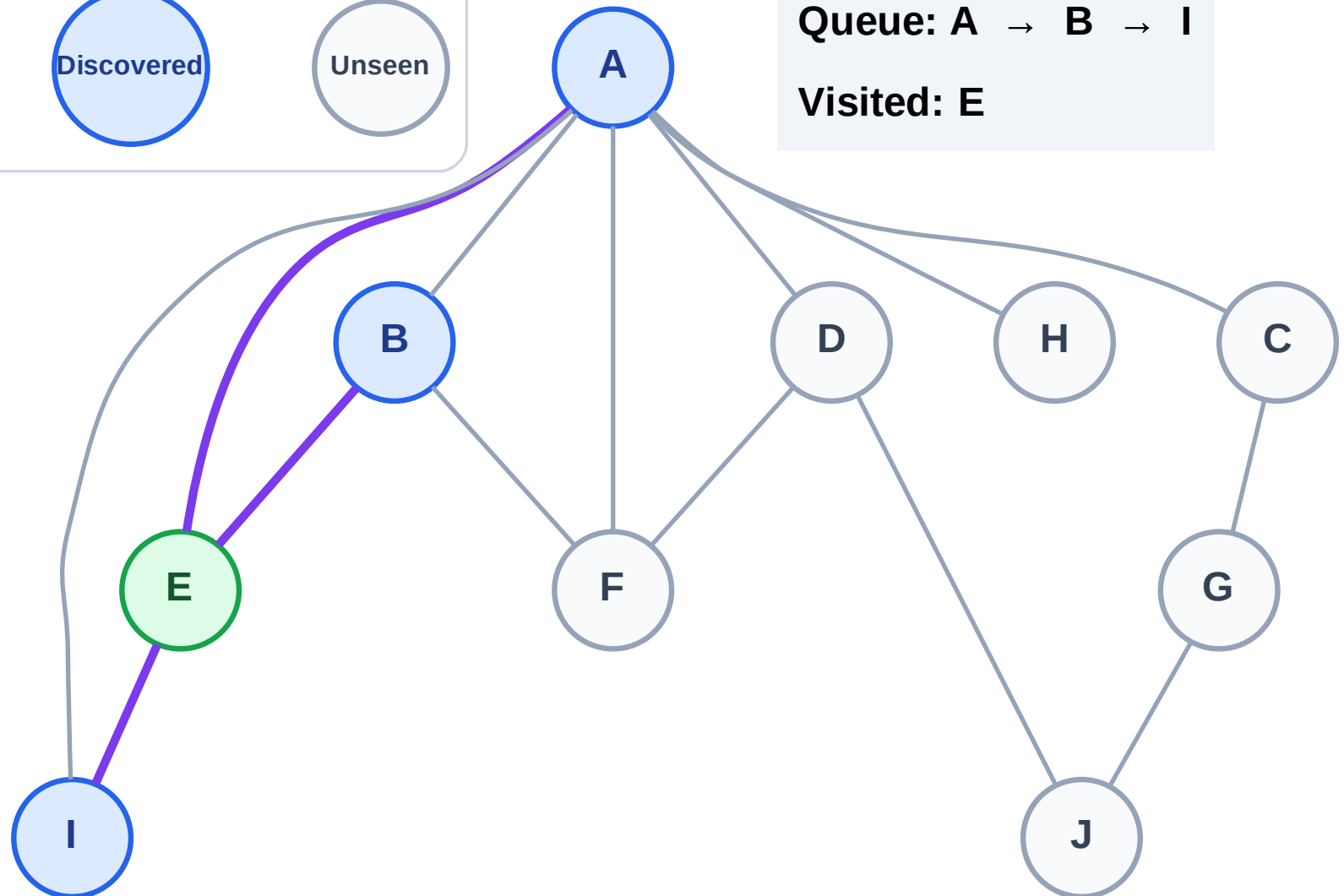
Step 3: Discover A, B, I from E

Legend



Queue: A → B → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

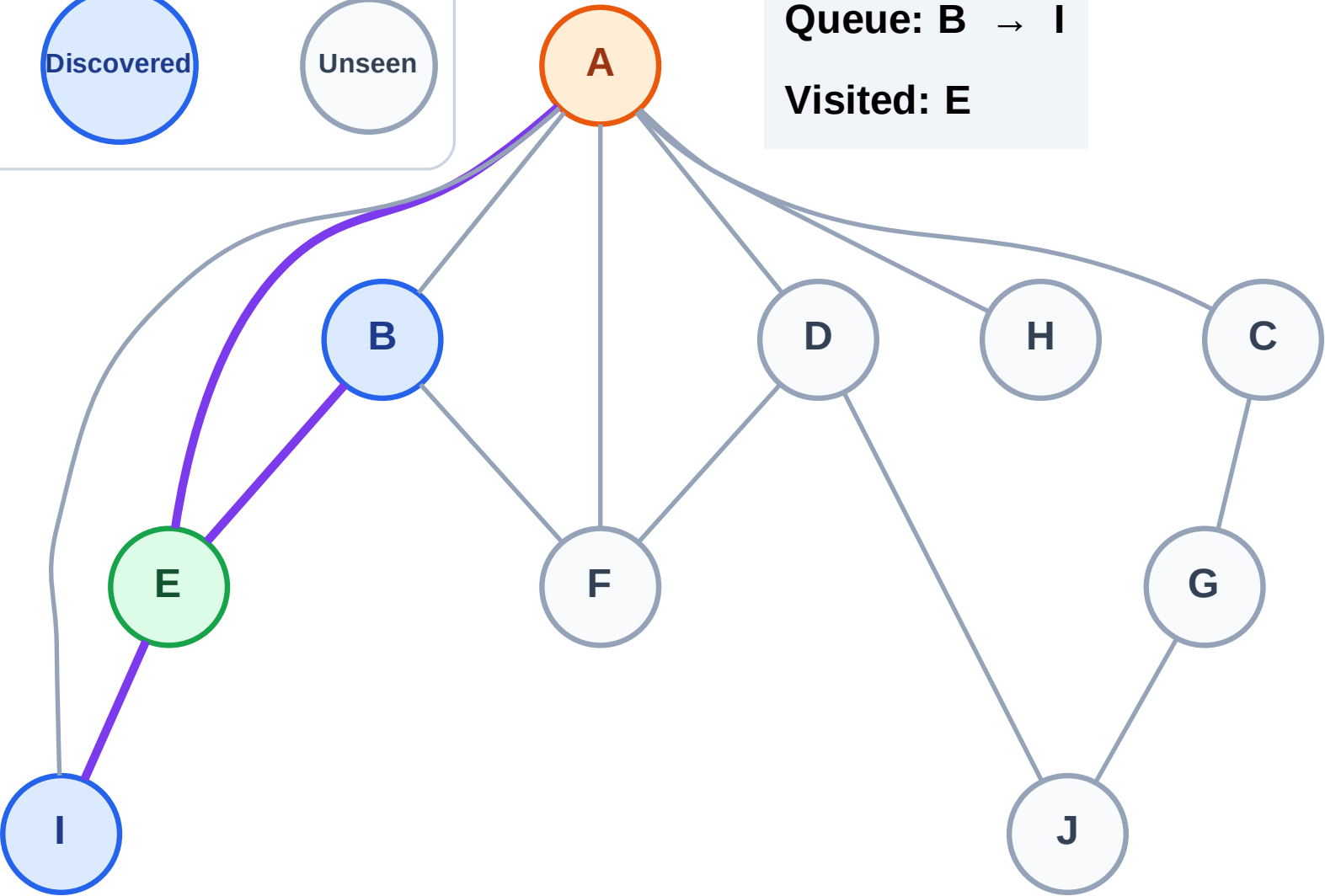
Processing

Discovered

Unseen

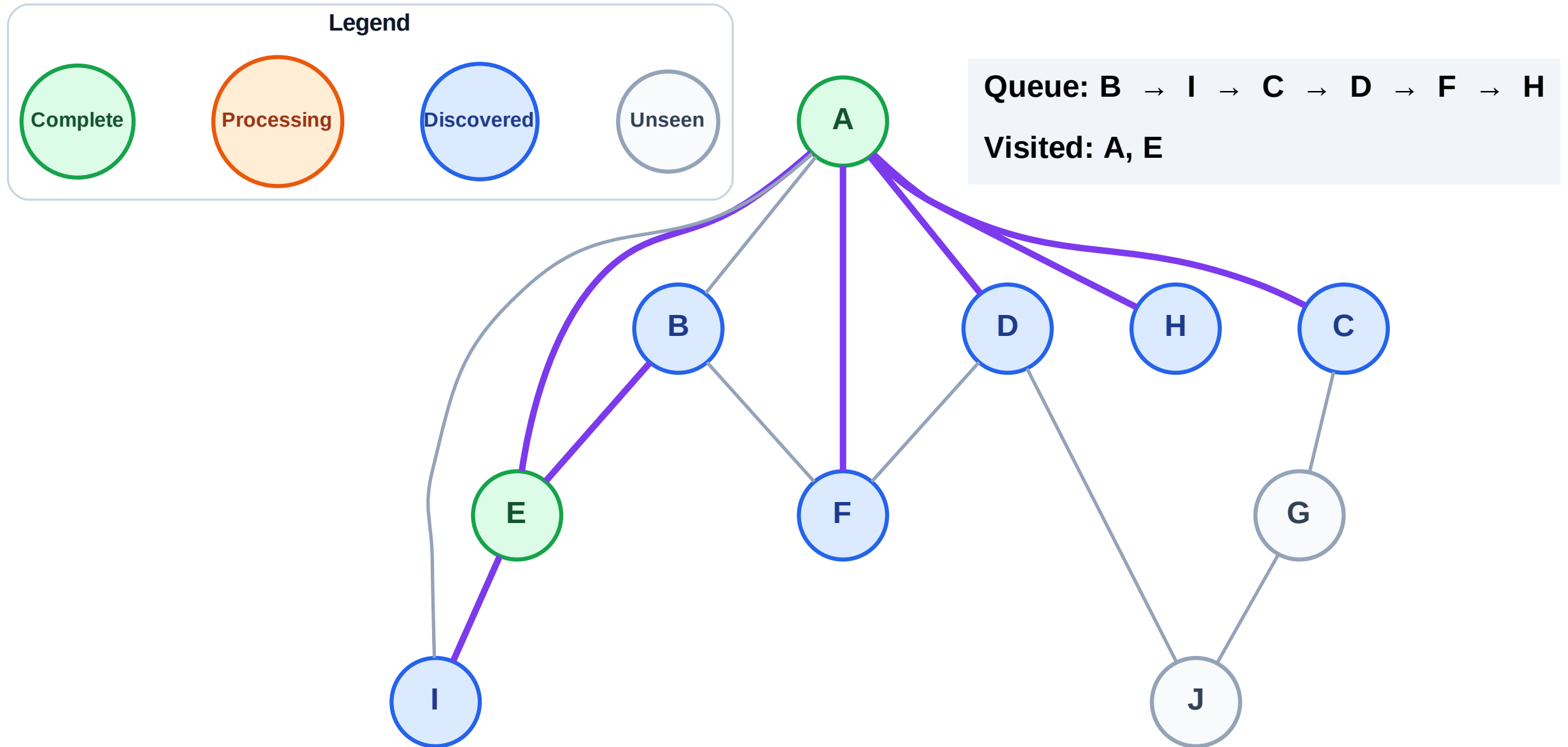
Queue: B → I

Visited: E



Step 5: Discover C, D, F, H from A

Step 5: Discover C, D, F, H from A



BFS Traversal

Step 6: Process node B

Legend

Complete

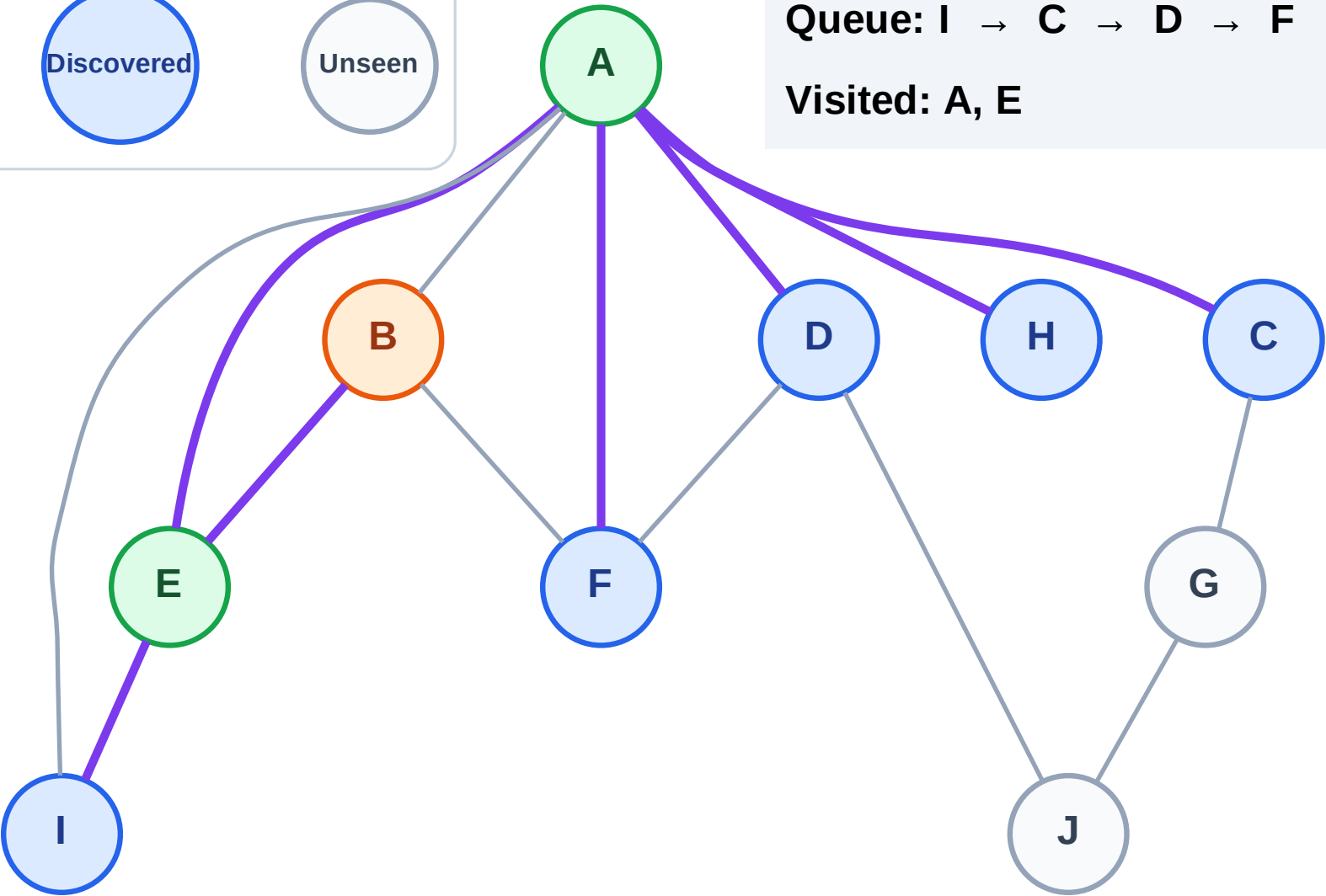
Processing

Discovered

Unseen

Queue: I → C → D → F → H

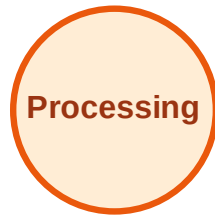
Visited: A, E



BFS Traversal

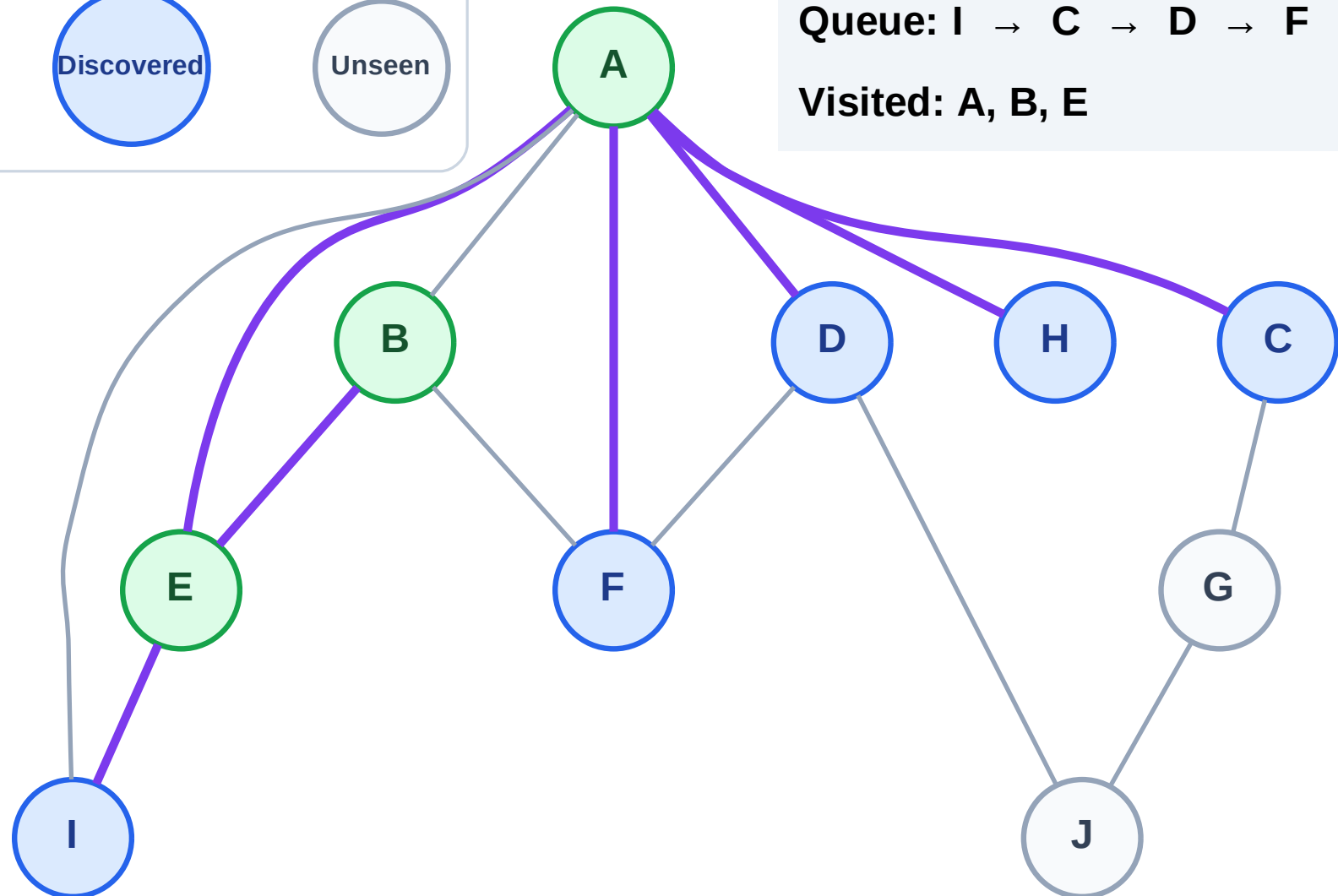
Step 7: No new neighbors from B

Legend



Queue: I → C → D → F → H

Visited: A, B, E



BFS Traversal

Step 8: Process node I

Legend

Complete

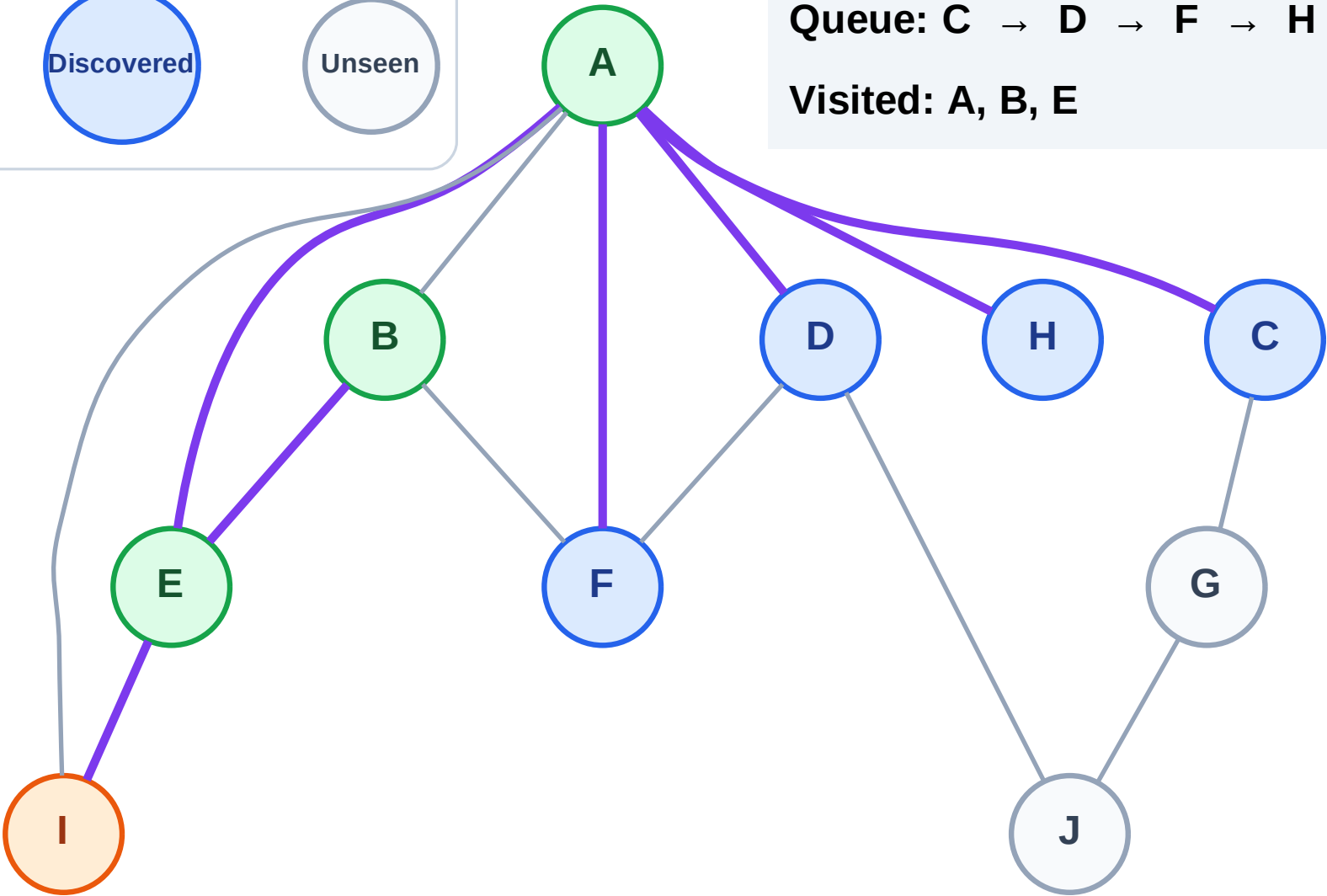
Processing

Discovered

Unseen

Queue: C → D → F → H

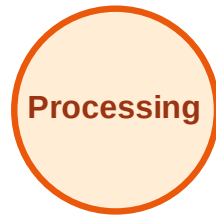
Visited: A, B, E



BFS Traversal

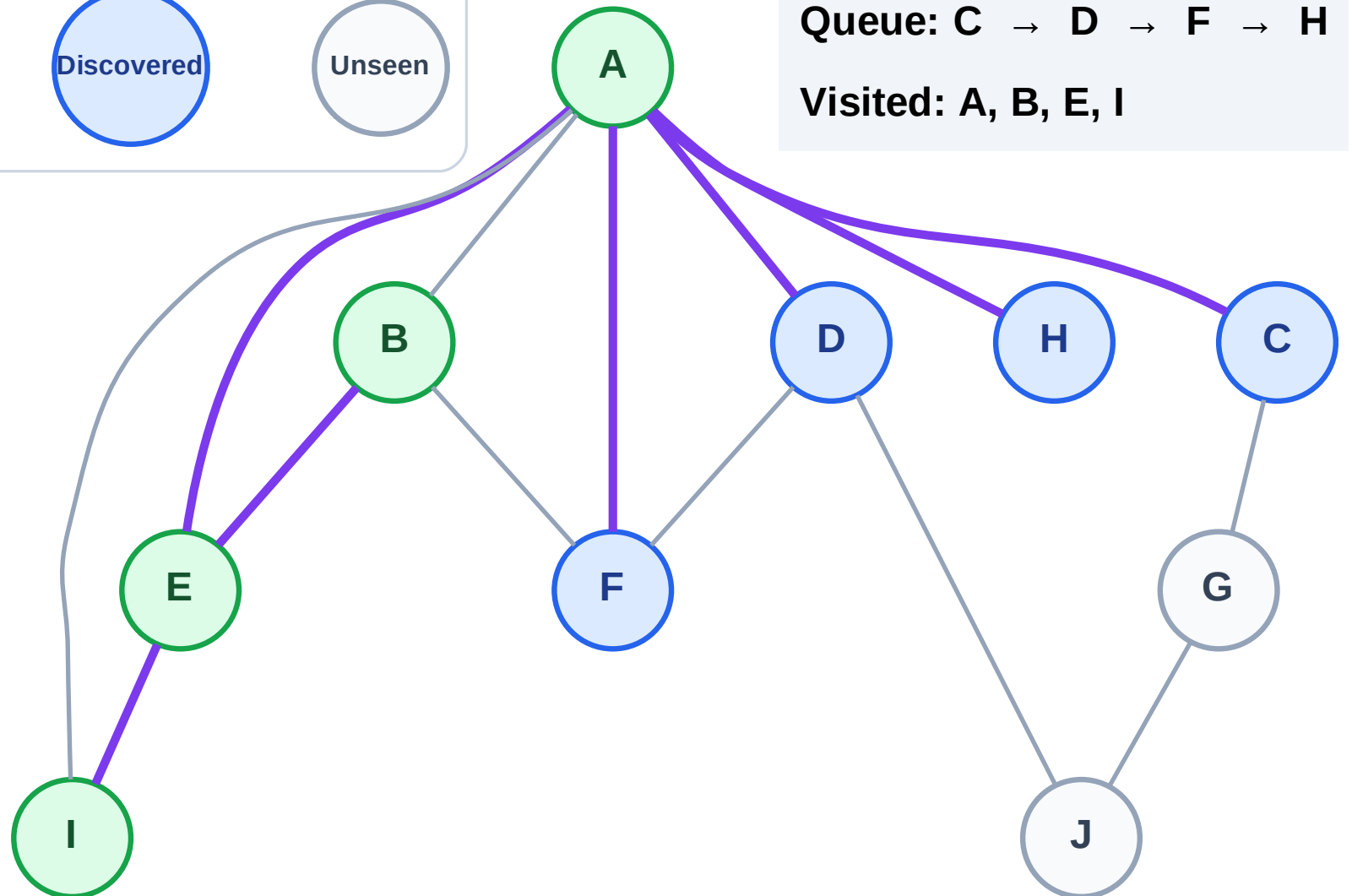
Step 9: No new neighbors from I

Legend



Queue: C → D → F → H

Visited: A, B, E, I



BFS Traversal

Step 10: Process node C

Legend

Complete

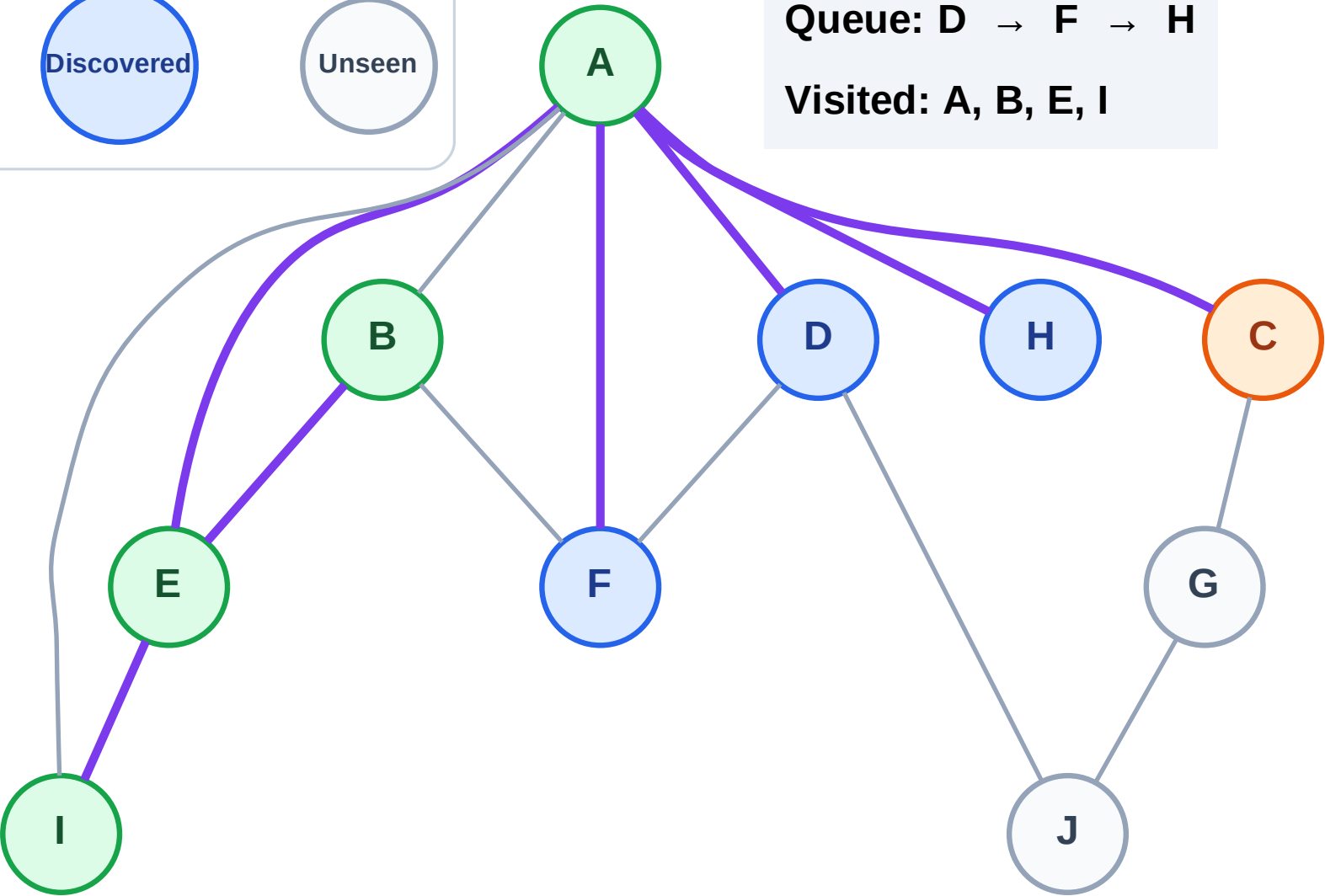
Processing

Discovered

Unseen

Queue: D → F → H

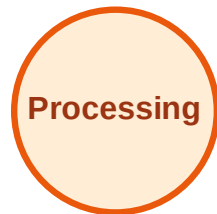
Visited: A, B, E, I



BFS Traversal

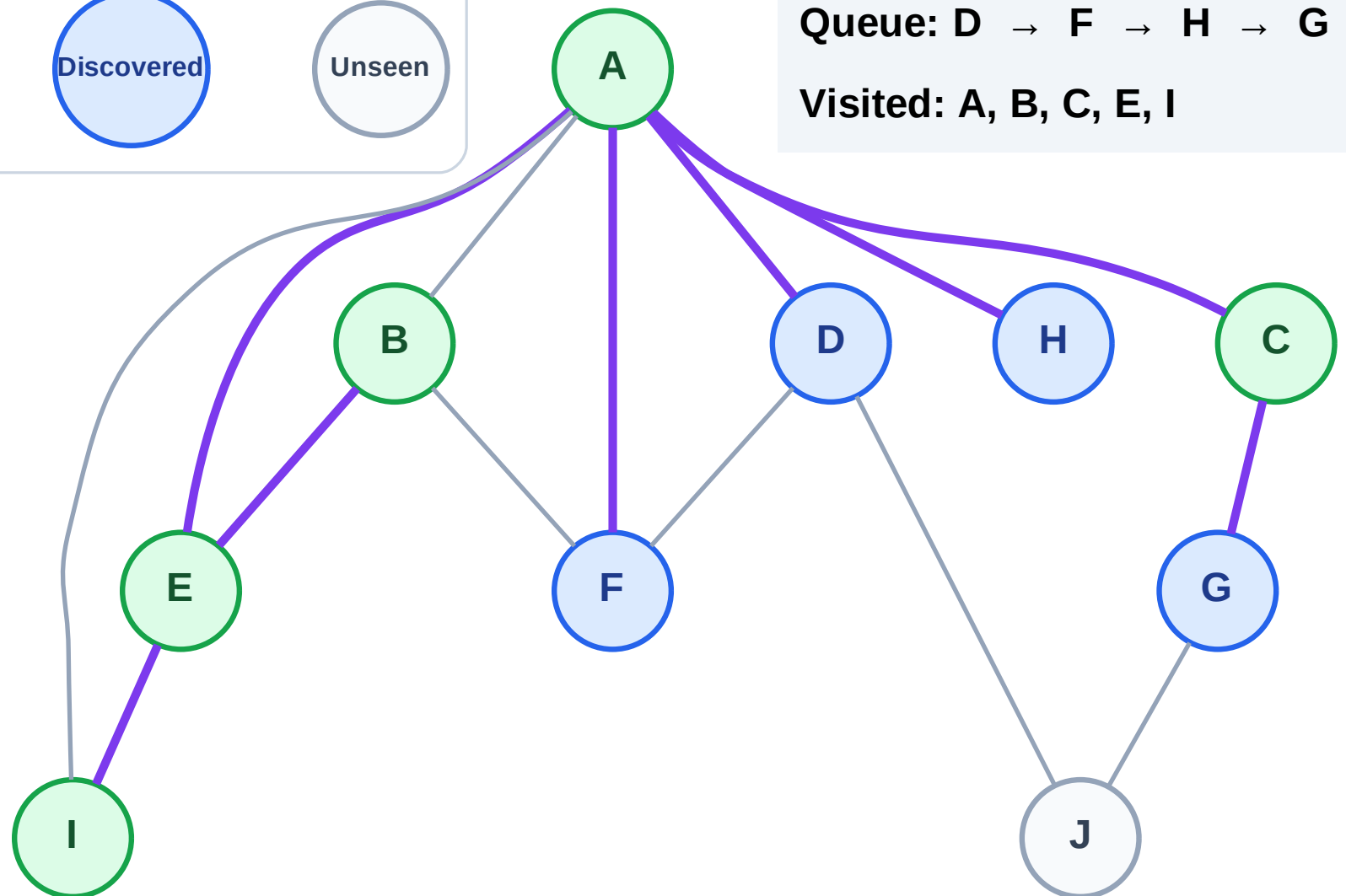
Step 11: Discover G from C

Legend



Queue: D → F → H → G

Visited: A, B, C, E, I



BFS Traversal

Step 12: Process node D

Legend

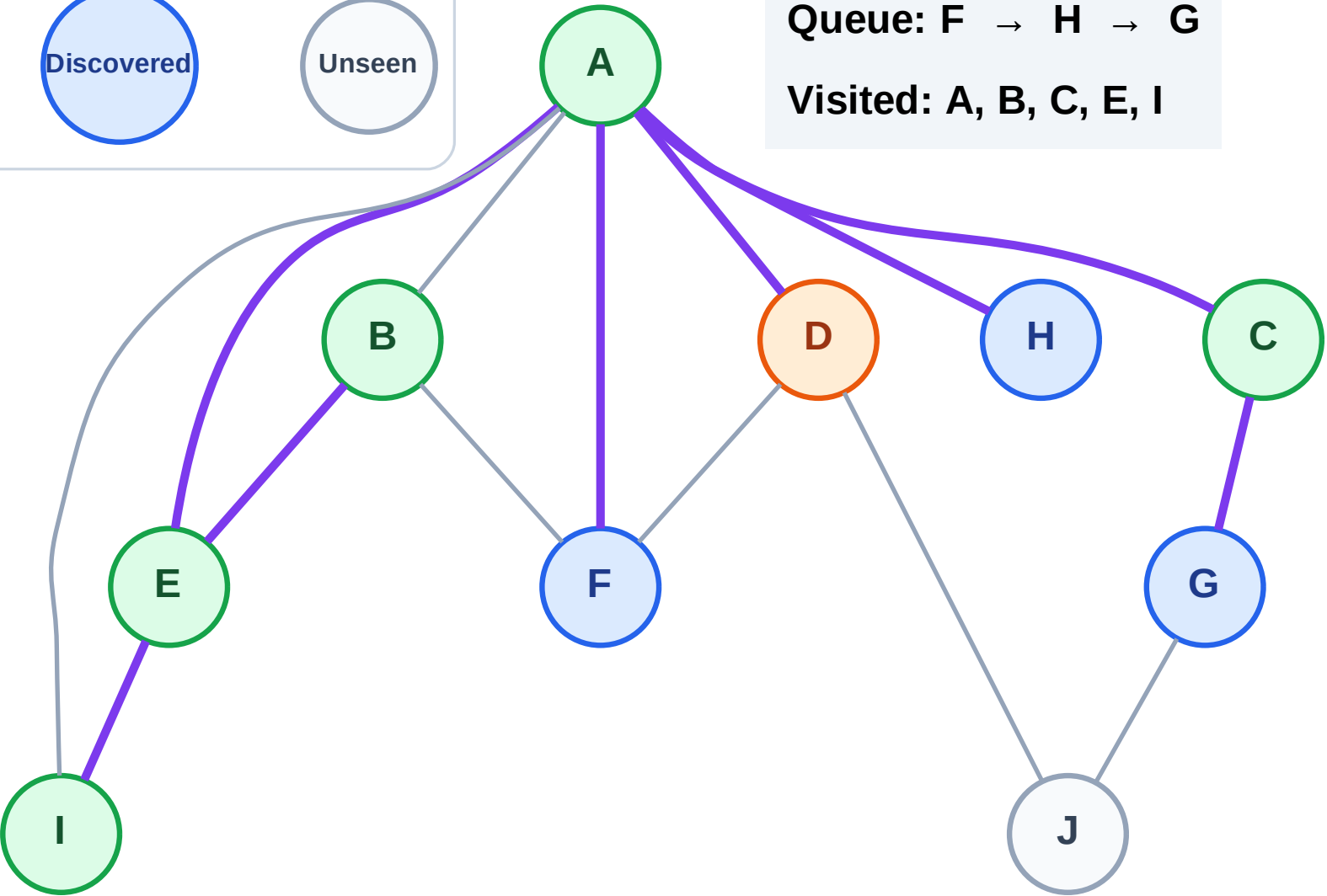
Complete

Processing

Discovered

Unseen

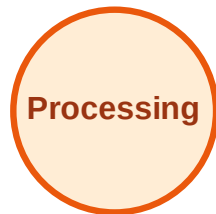
Queue: F → H → G
Visited: A, B, C, E, I



BFS Traversal

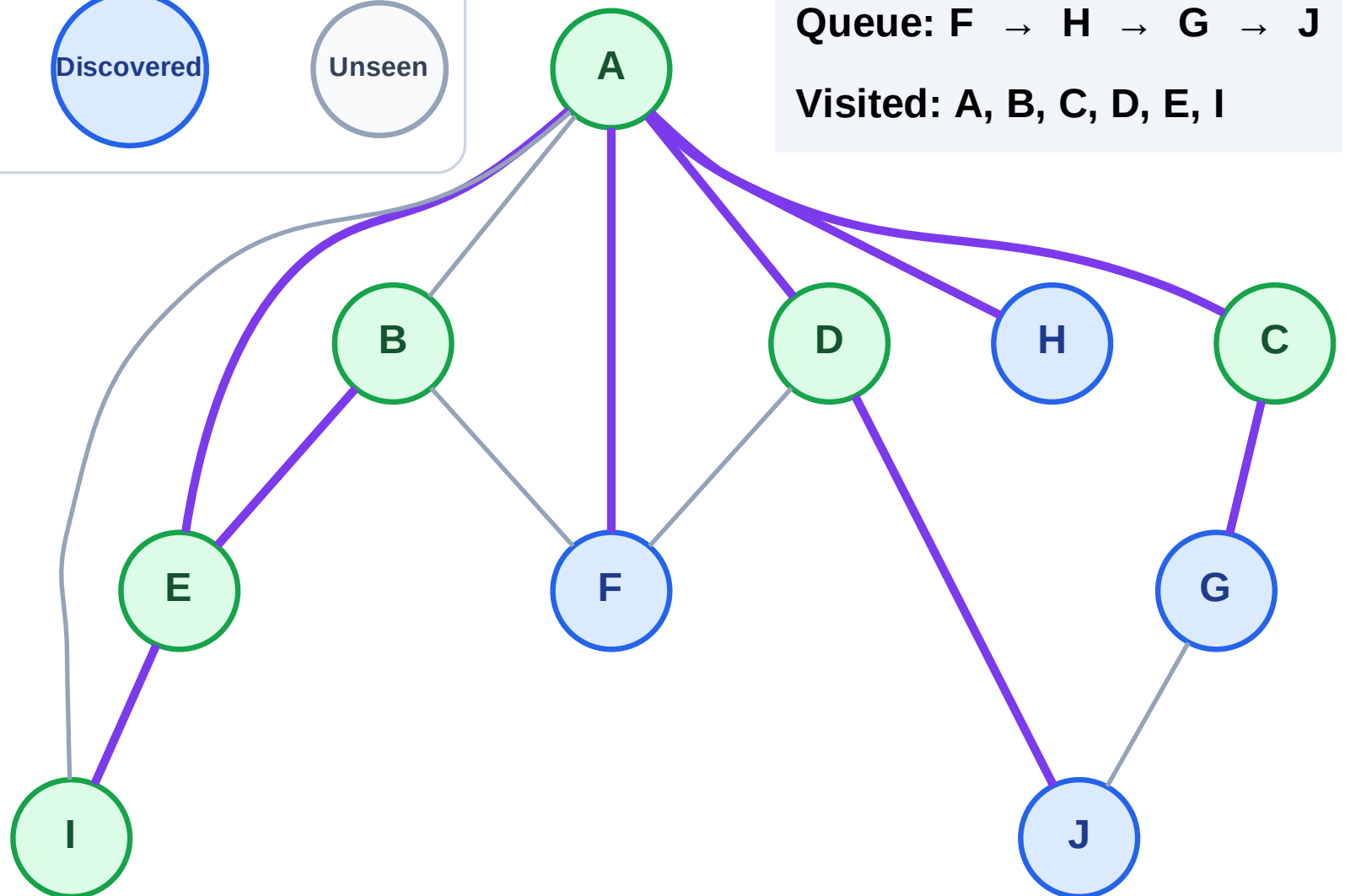
Step 13: Discover J from D

Legend



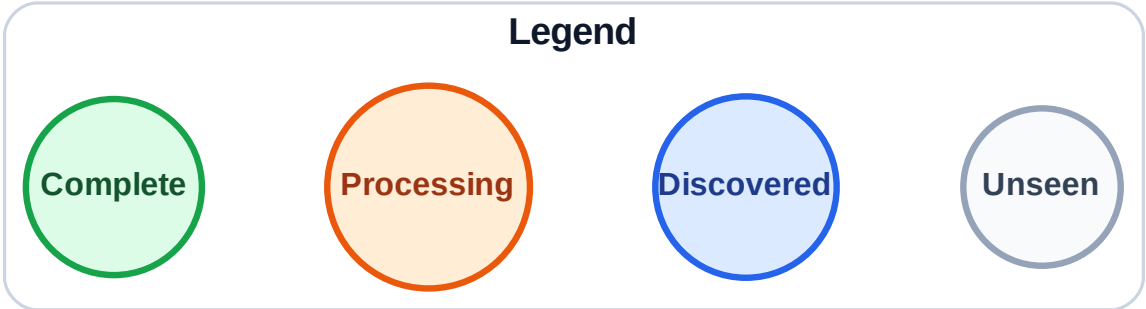
Queue: F → H → G → J

Visited: A, B, C, D, E, I

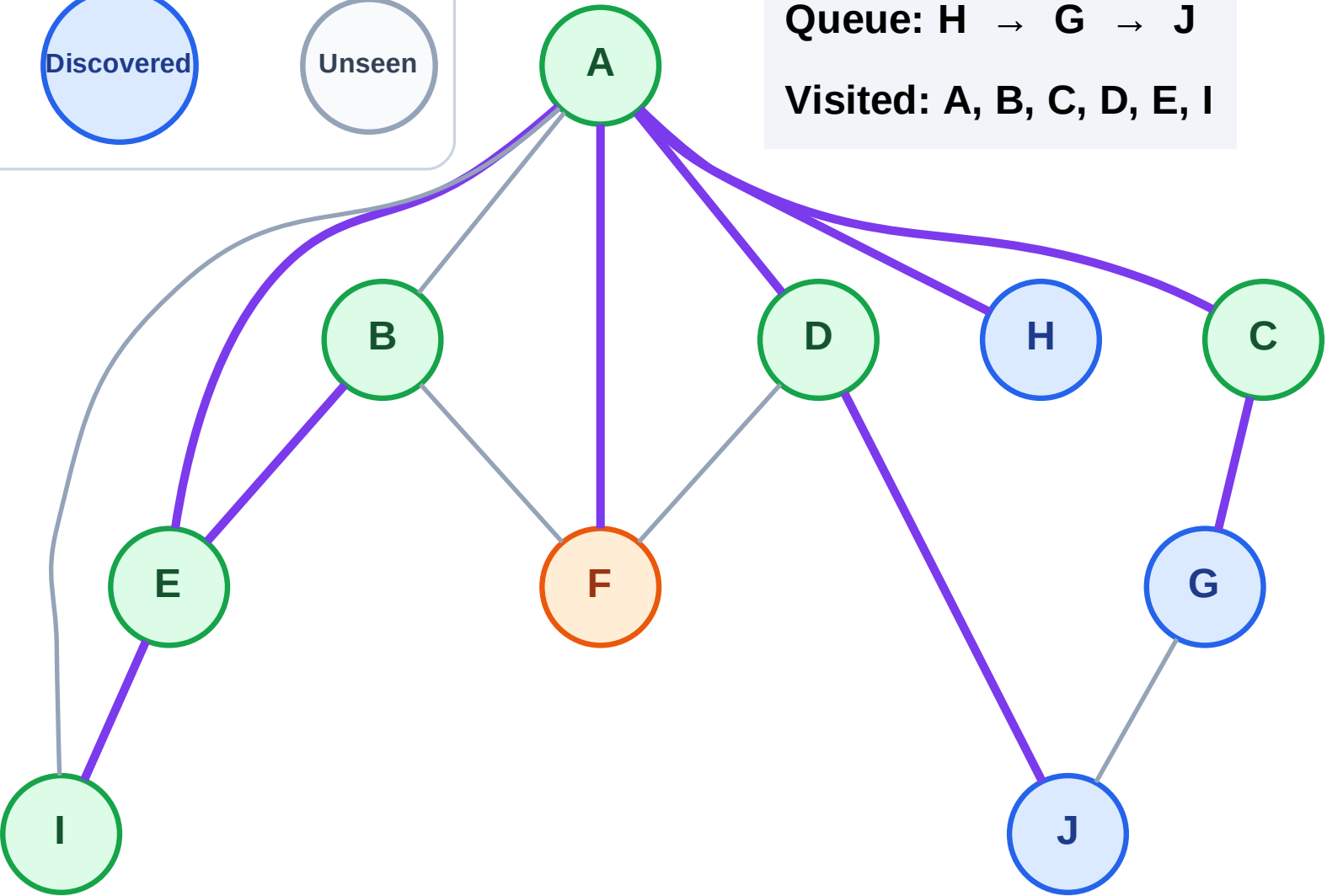


BFS Traversal

Step 14: Process node F



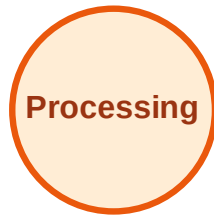
Queue: H → G → J
Visited: A, B, C, D, E, I



BFS Traversal

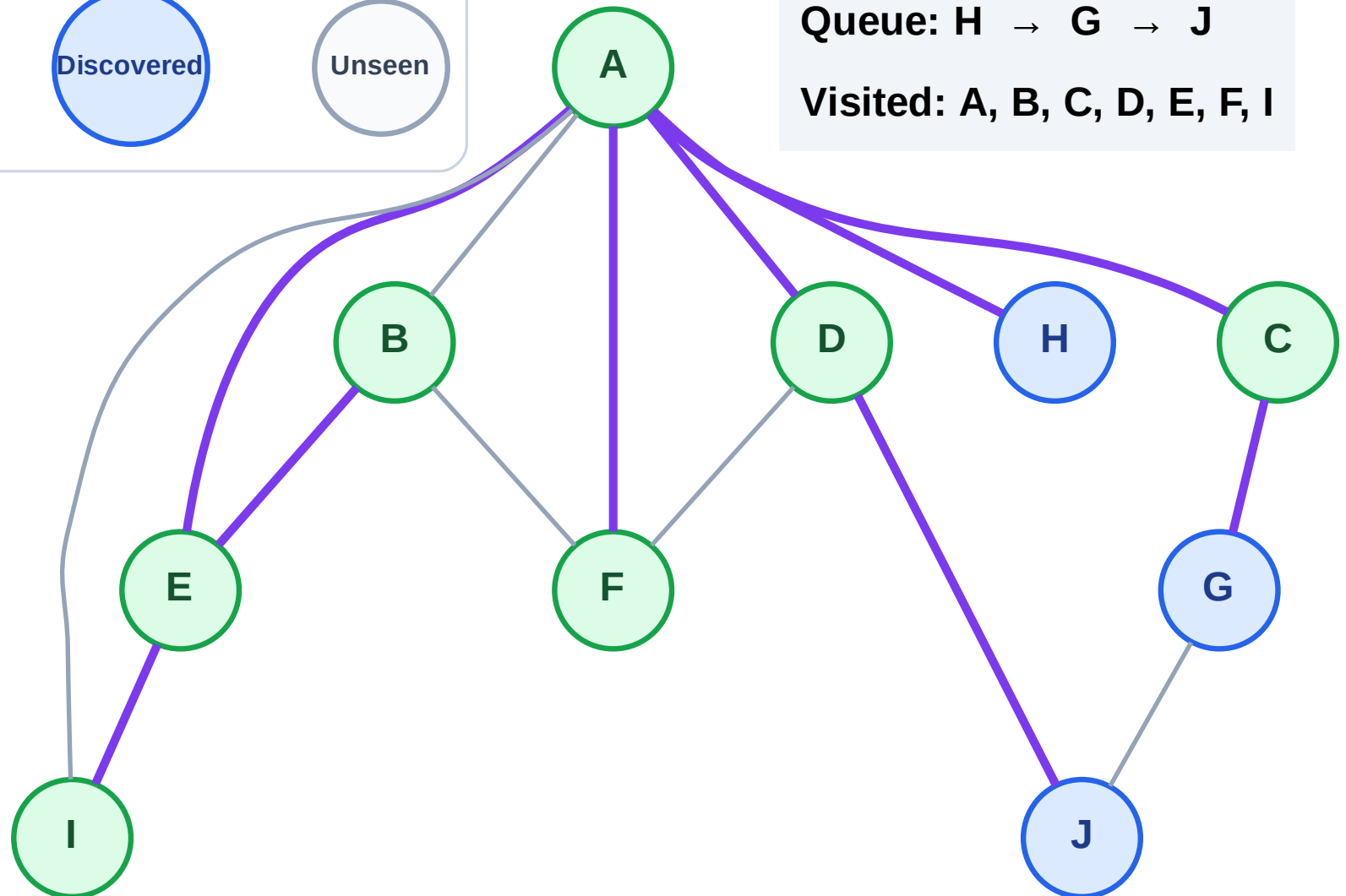
Step 15: No new neighbors from F

Legend



Queue: H → G → J

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node H

Legend

Complete

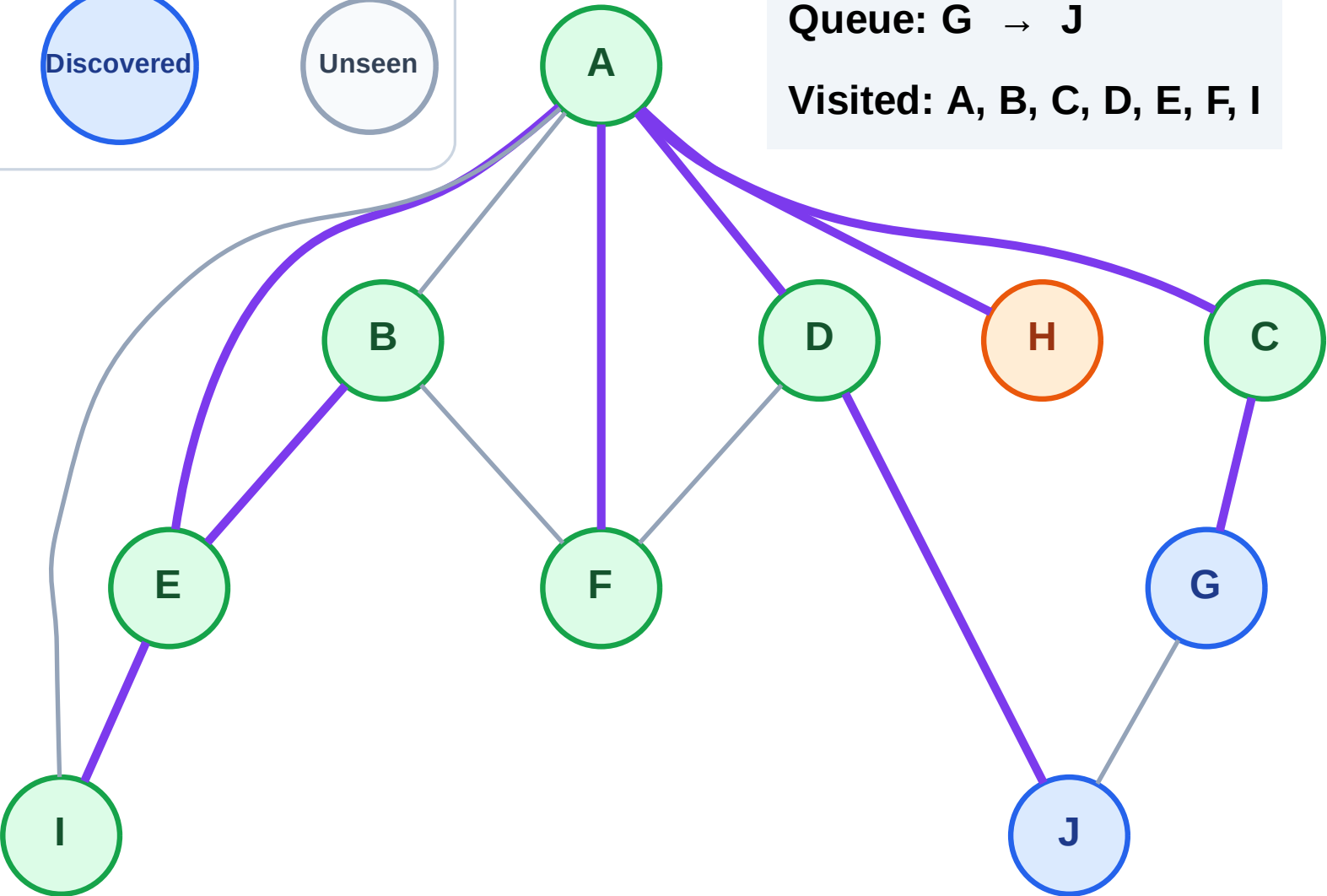
Processing

Discovered

Unseen

Queue: G → J

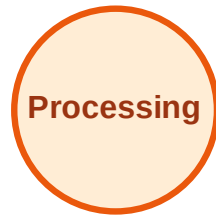
Visited: A, B, C, D, E, F, I



BFS Traversal

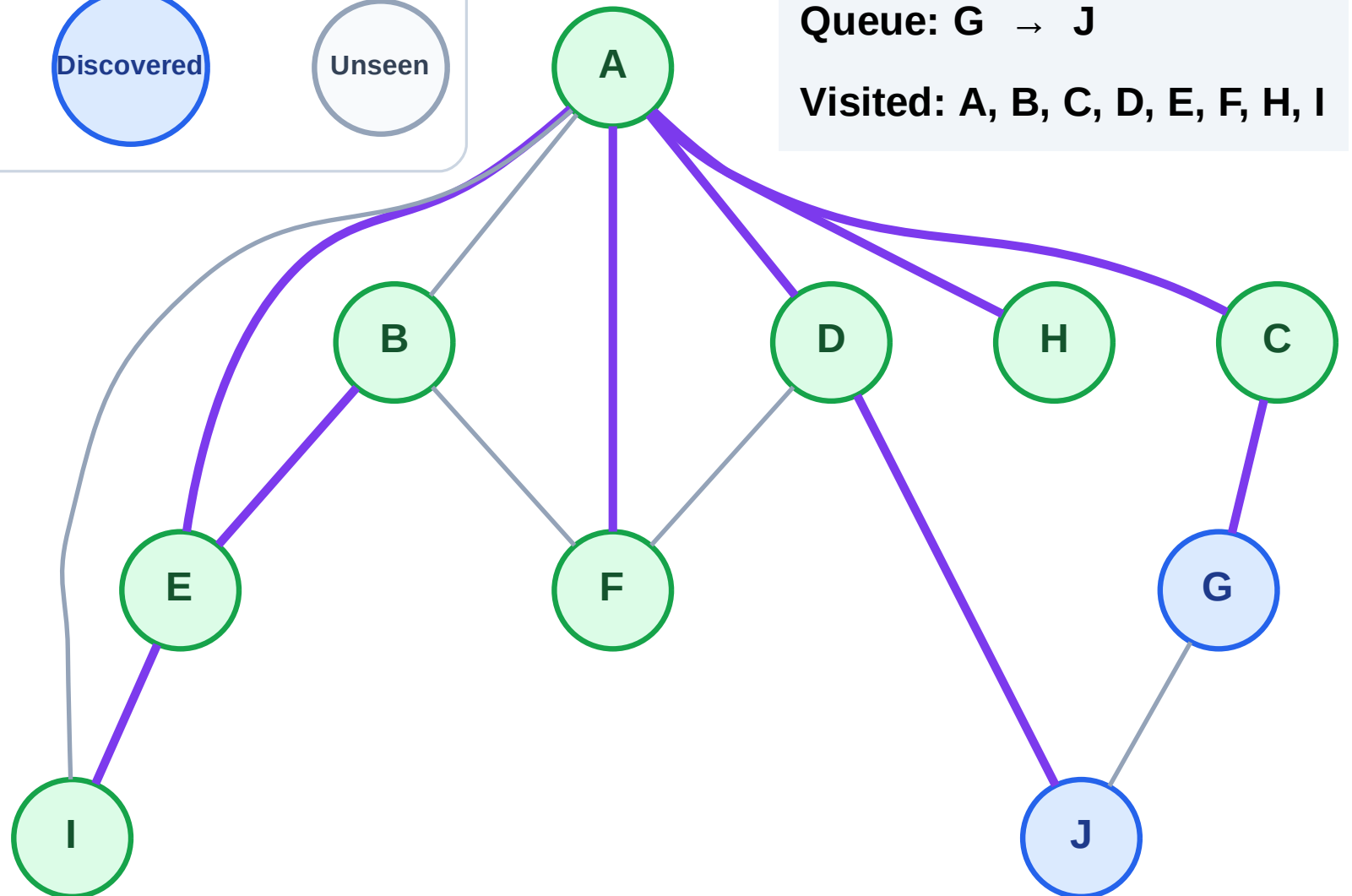
Step 17: No new neighbors from H

Legend



Queue: G → J

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 18: Process node G

Legend

Complete

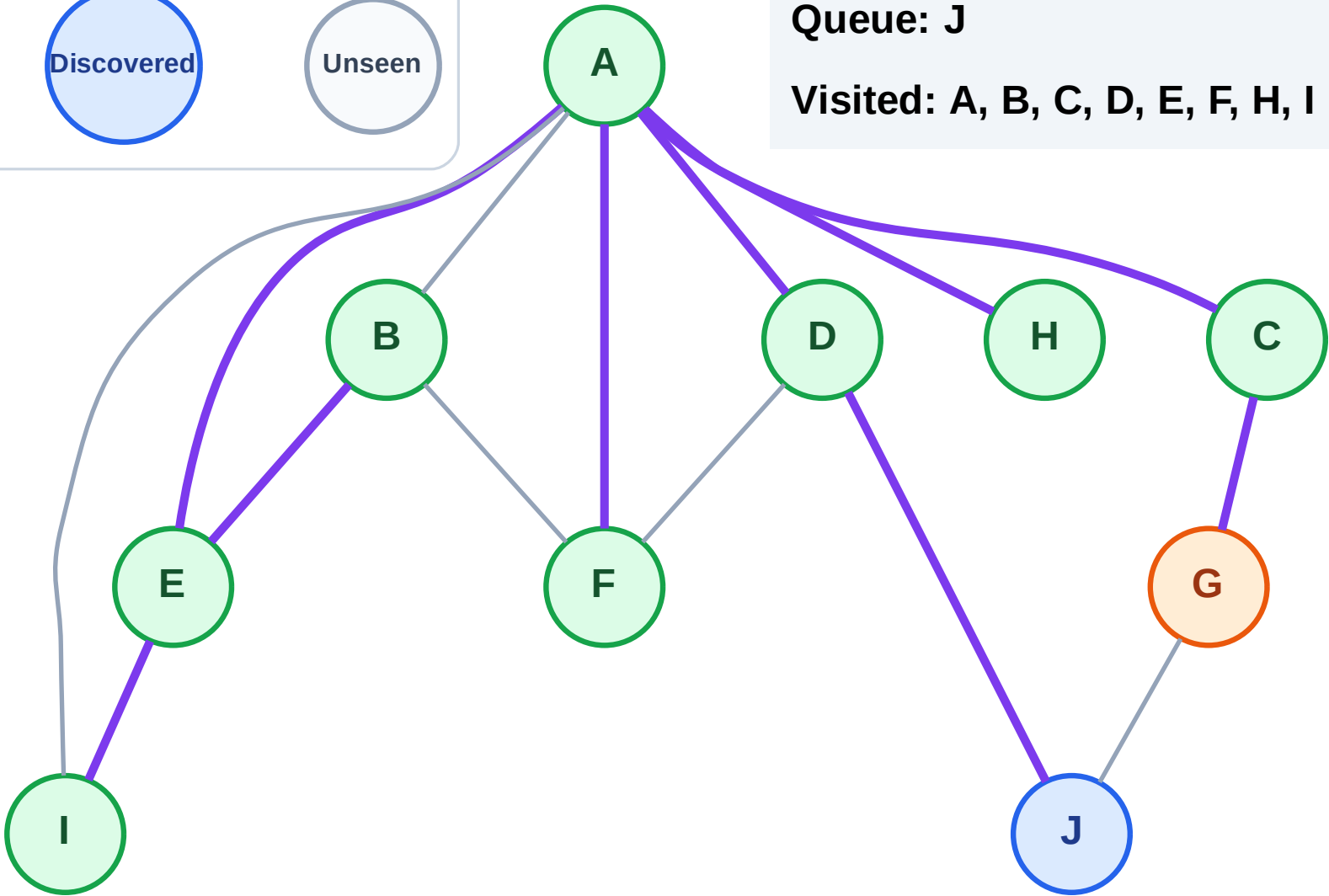
Processing

Discovered

Unseen

Queue: J

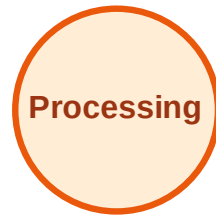
Visited: A, B, C, D, E, F, H, I



BFS Traversal

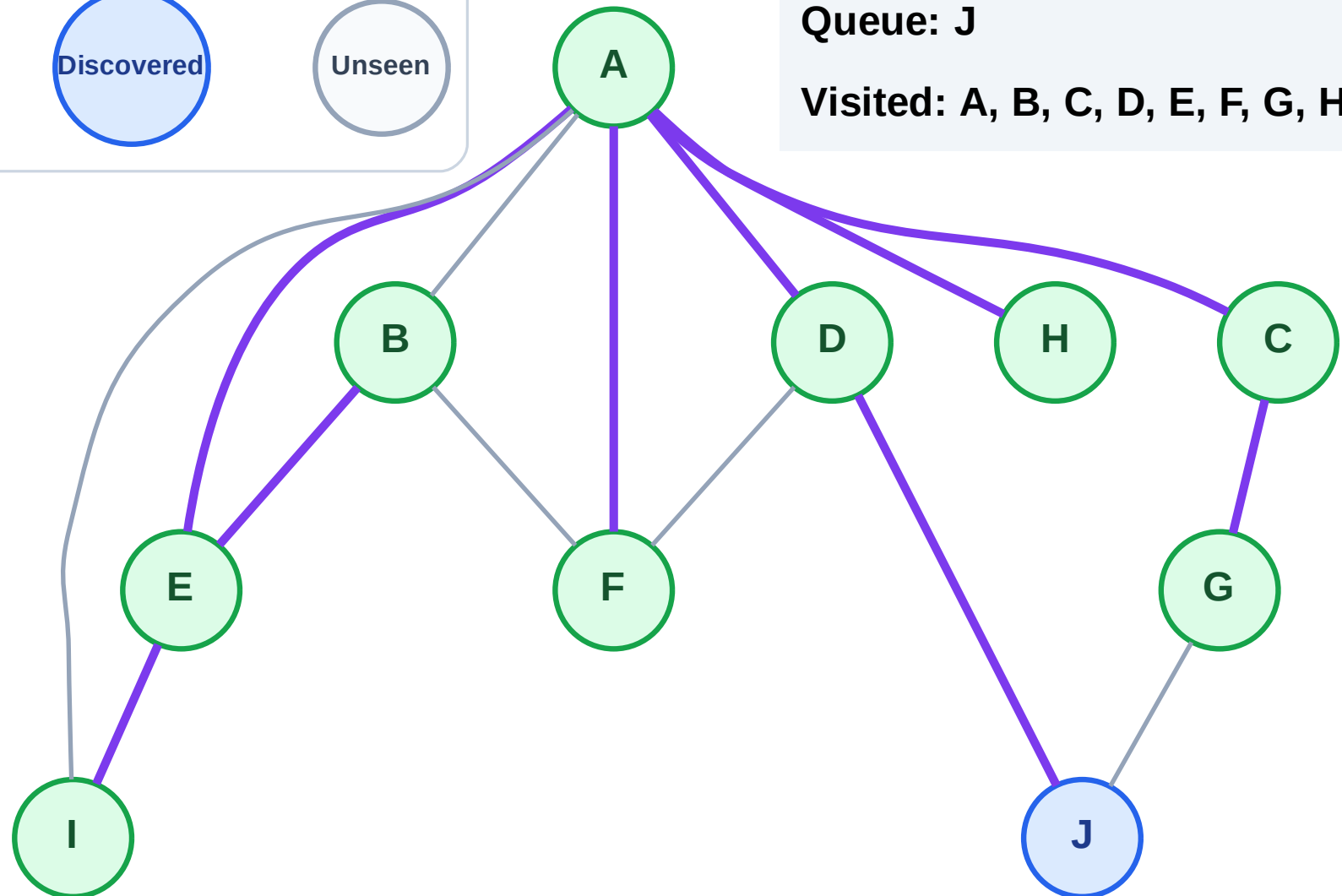
Step 19: No new neighbors from G

Legend



Queue: J

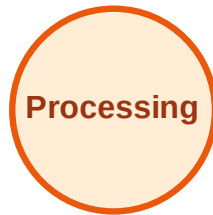
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

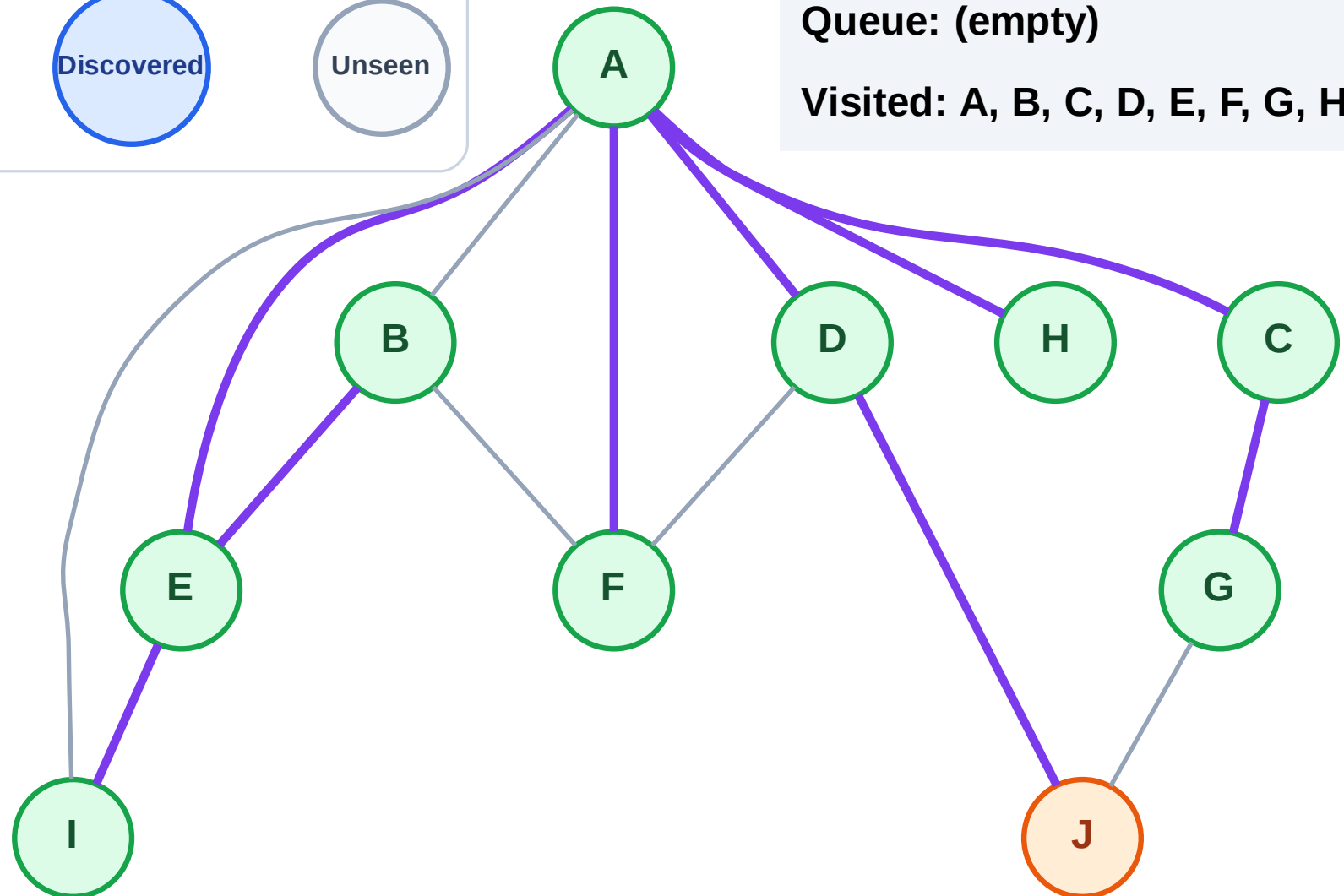
Step 20: Process node J

Legend



Queue: (empty)

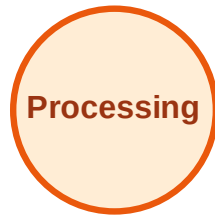
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

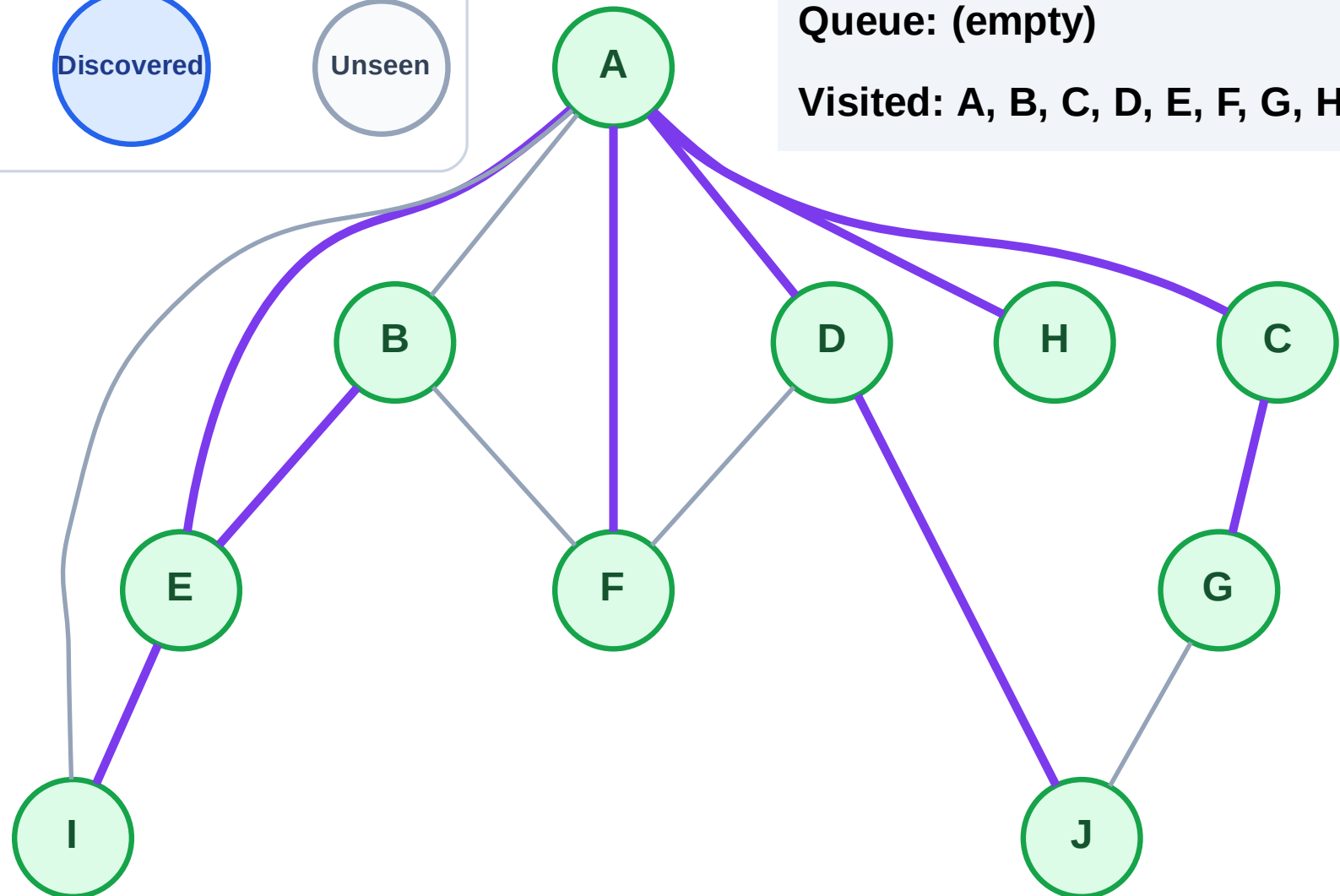
Step 21: No new neighbors from J

Legend



Queue: (empty)

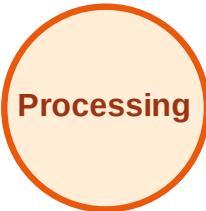
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

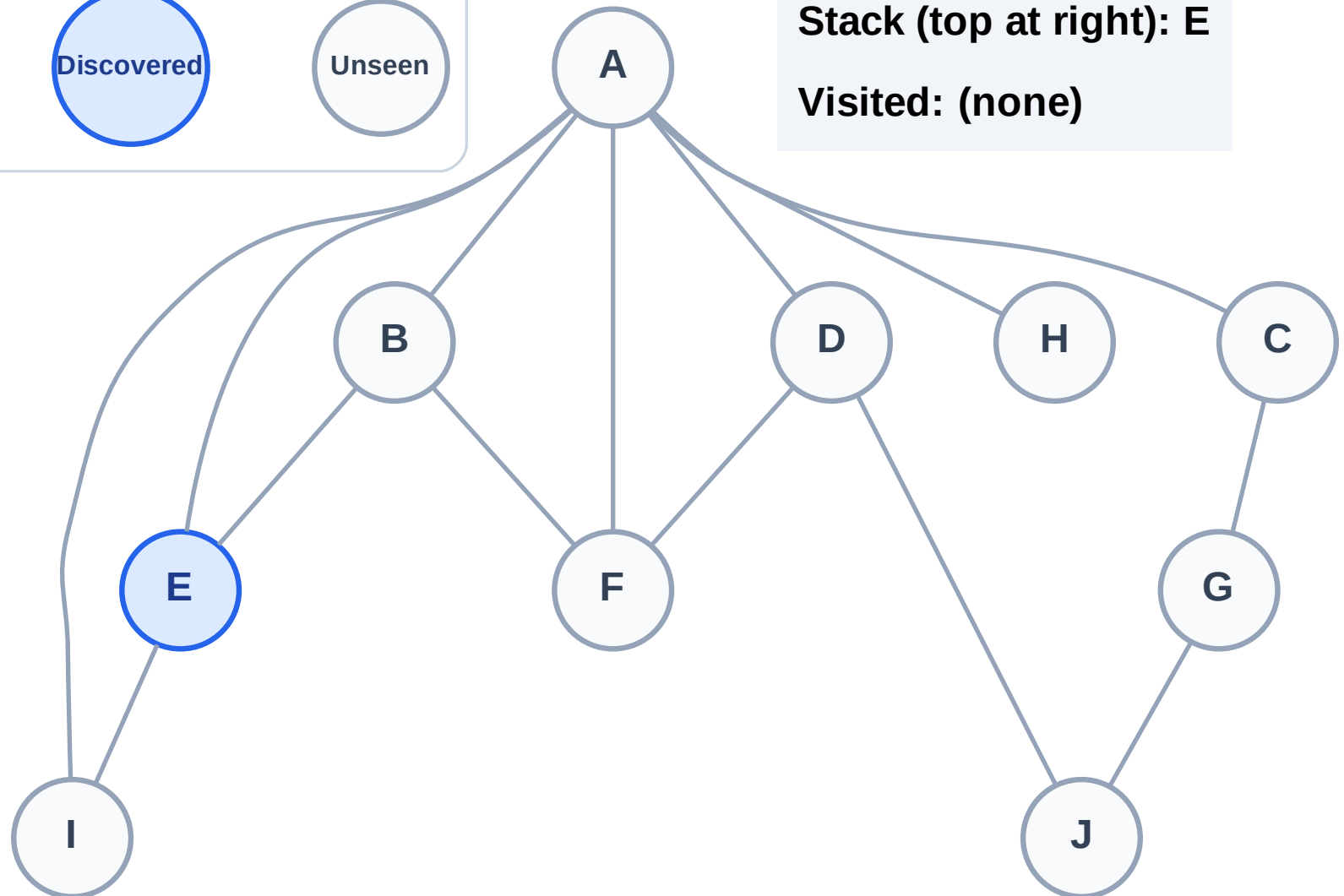
Step 1: Start at E

Legend



Stack (top at right): E

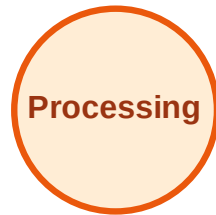
Visited: (none)



DFS Traversal

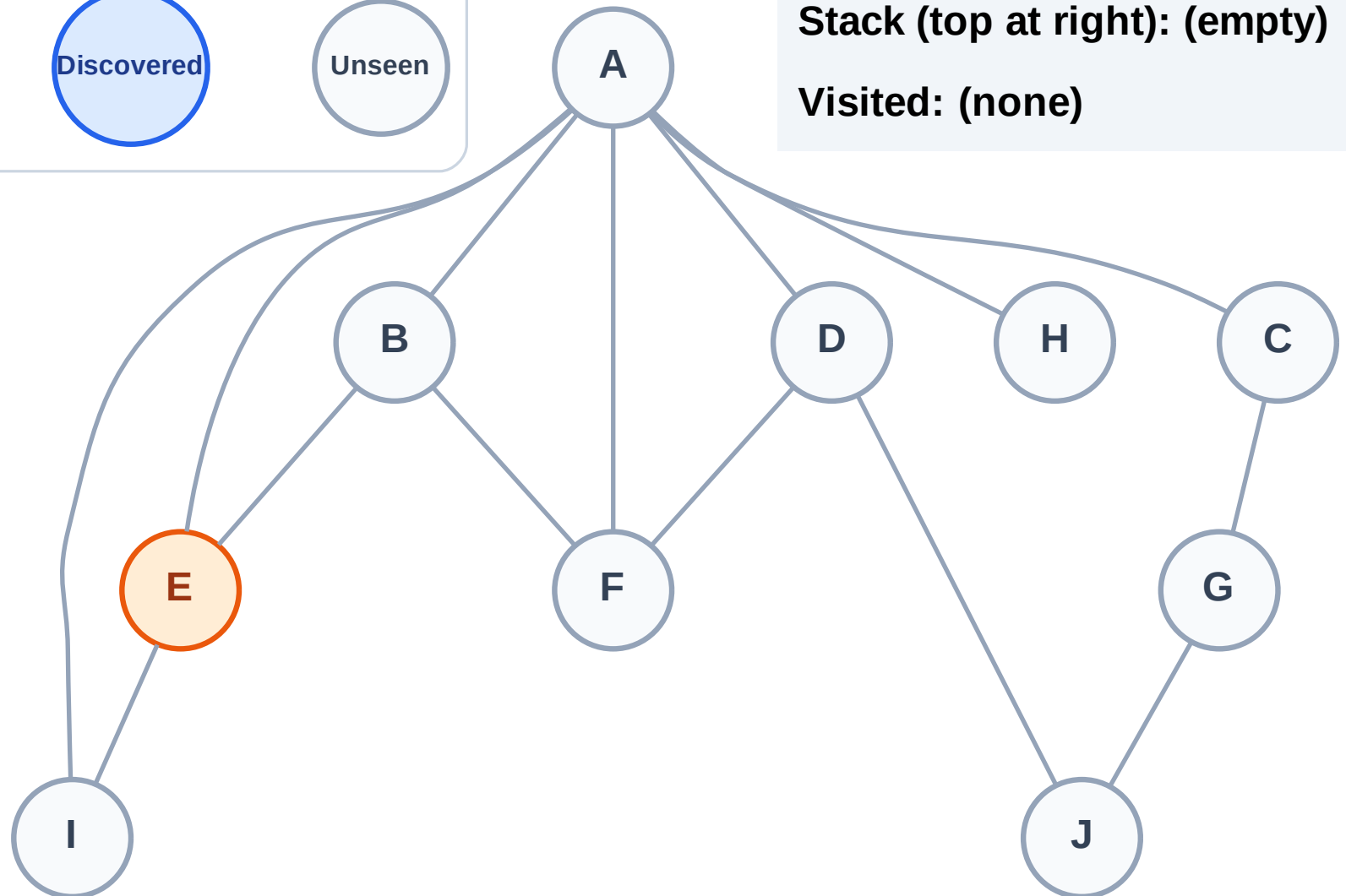
Step 2: Process node E

Legend



Stack (top at right): (empty)

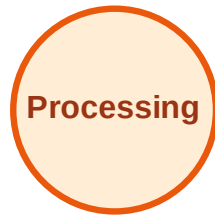
Visited: (none)



DFS Traversal

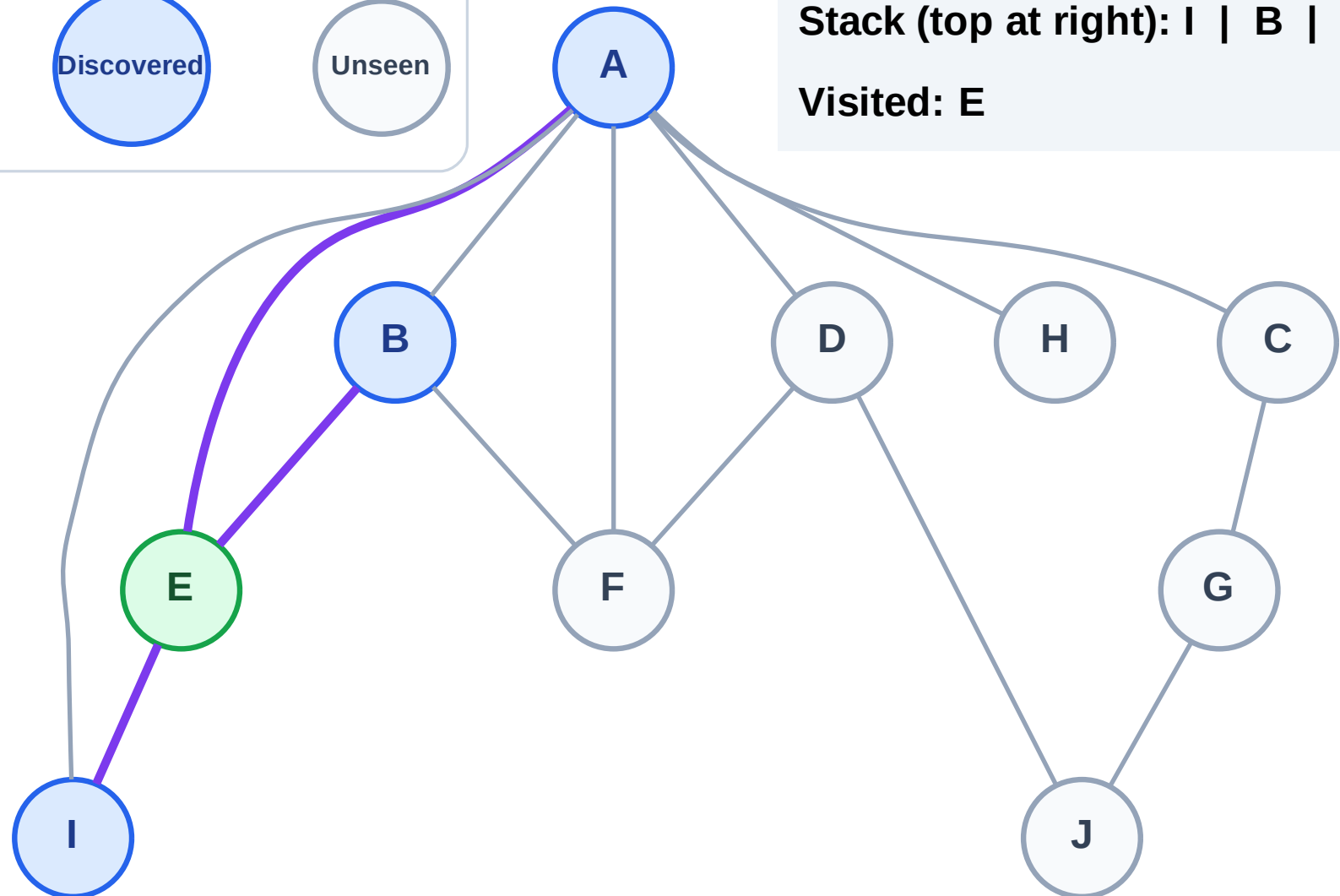
Step 3: Discover I, B, A from E

Legend



Stack (top at right): I | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

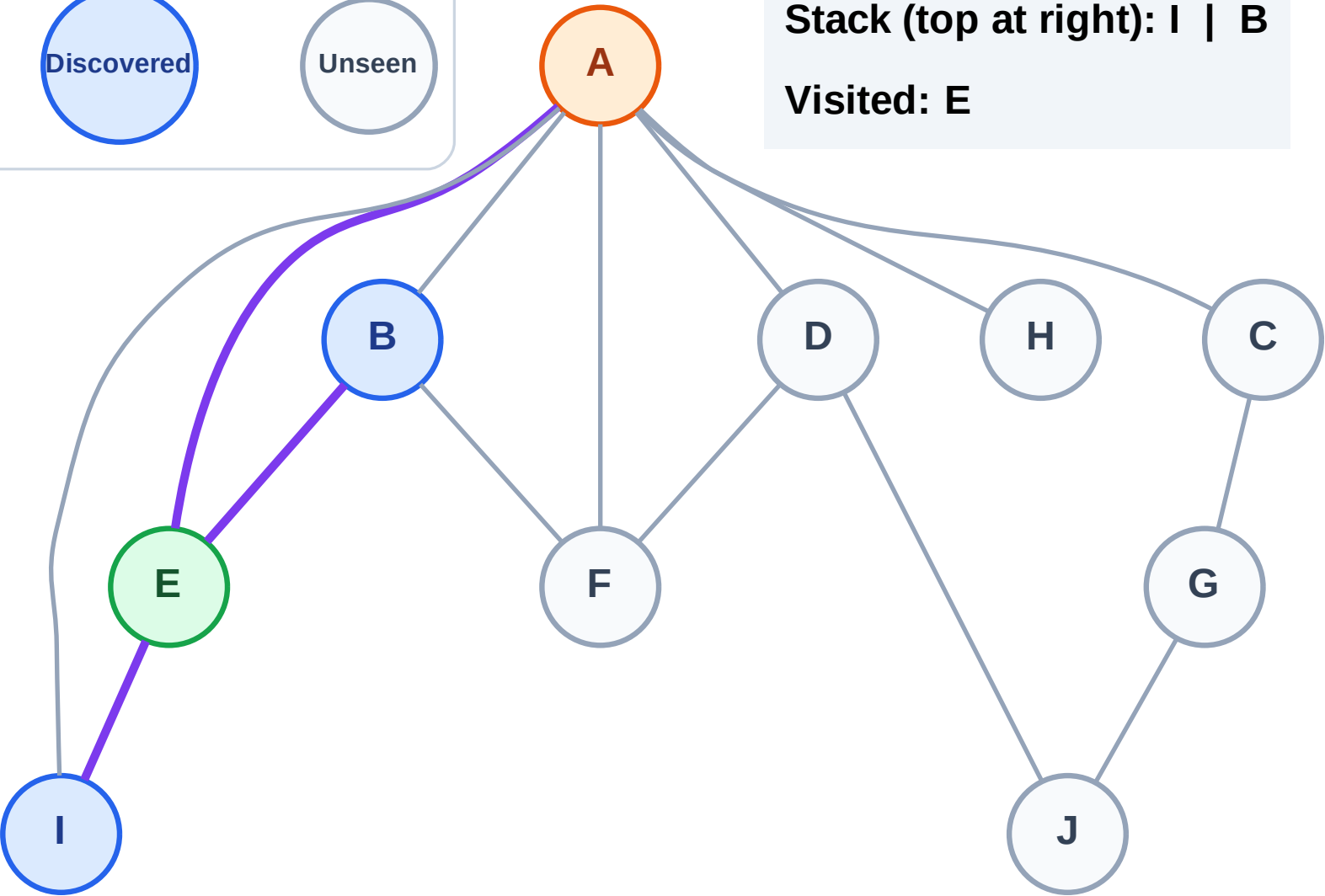
Complete

Processing

Discovered

Unseen

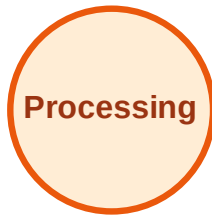
Stack (top at right): I | B
Visited: E



DFS Traversal

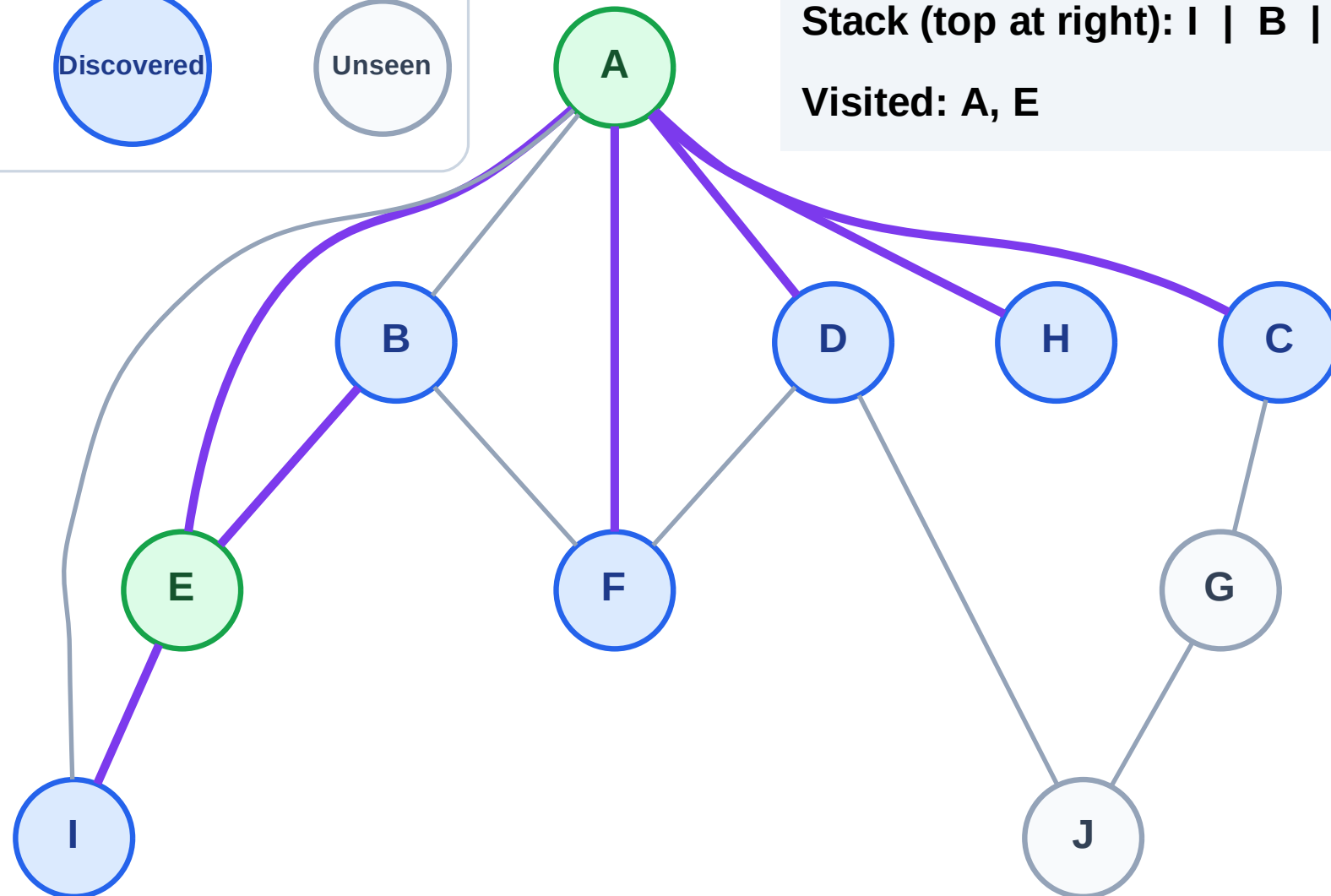
Step 5: Discover H, F, D, C from A

Legend



Stack (top at right): I | B | H | F | D | C

Visited: A, E



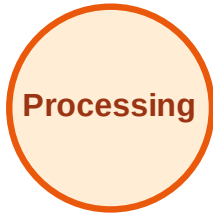
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



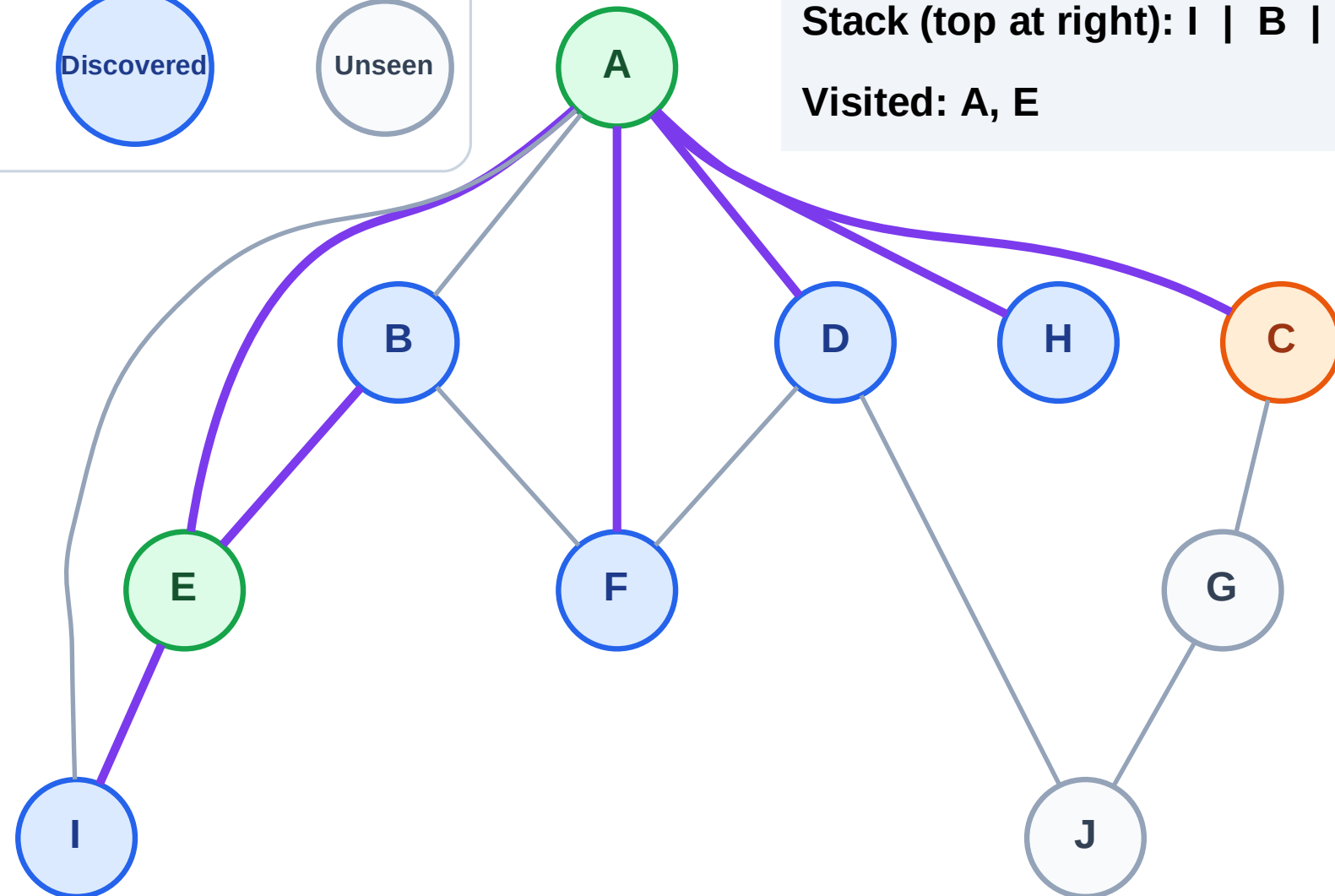
Discovered



Unseen

Stack (top at right): I | B | H | F | D

Visited: A, E



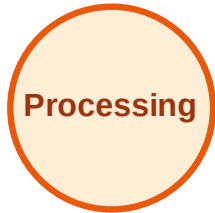
DFS Traversal

Step 7: Discover G from C

Legend



Complete



Processing



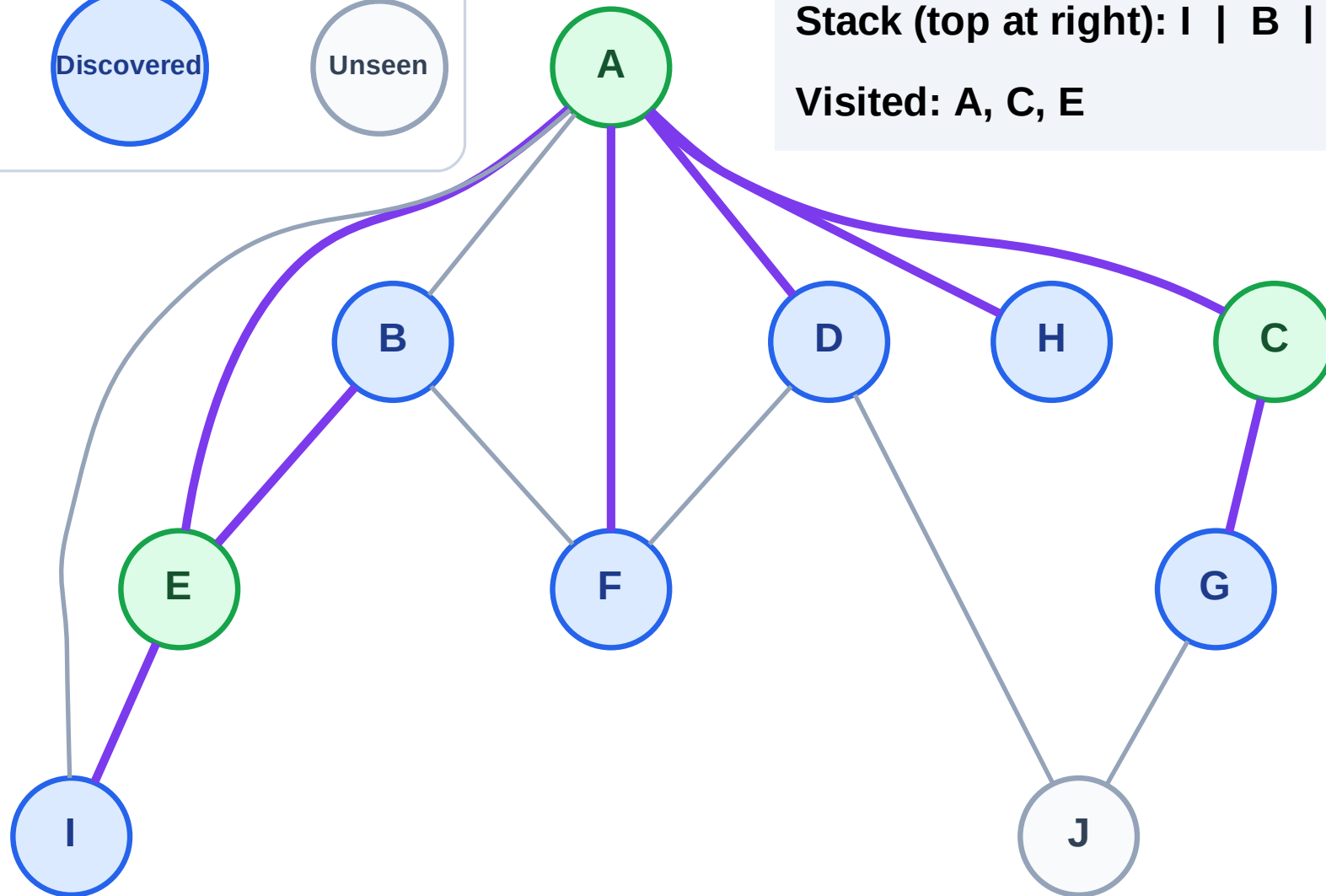
Discovered



Unseen

Stack (top at right): I | B | H | F | D | G

Visited: A, C, E



DFS Traversal

Step 8: Process node G

Legend

Complete

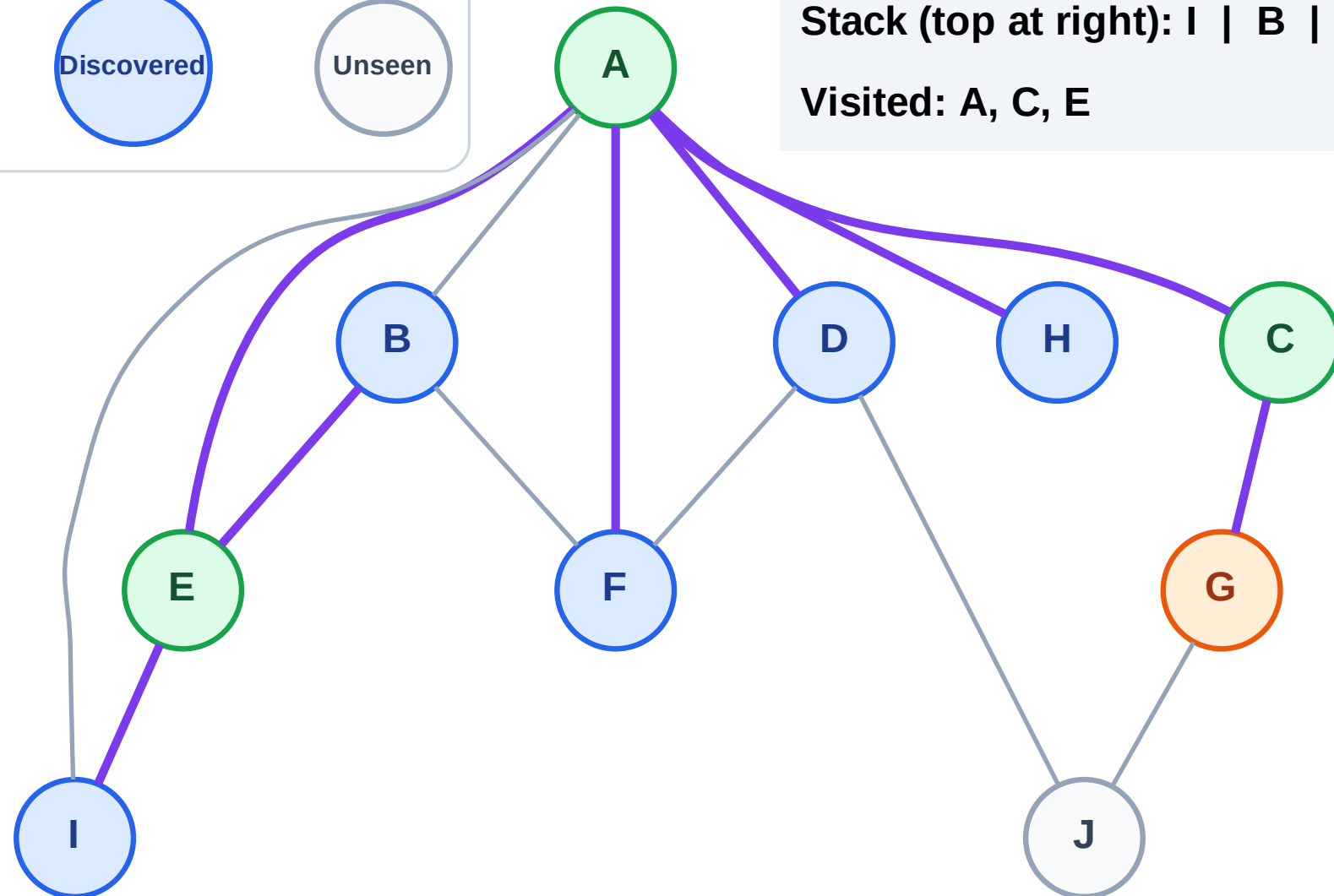
Processing

Discovered

Unseen

Stack (top at right): I | B | H | F | D

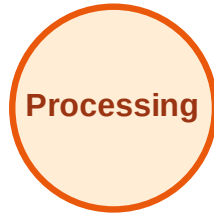
Visited: A, C, E



DFS Traversal

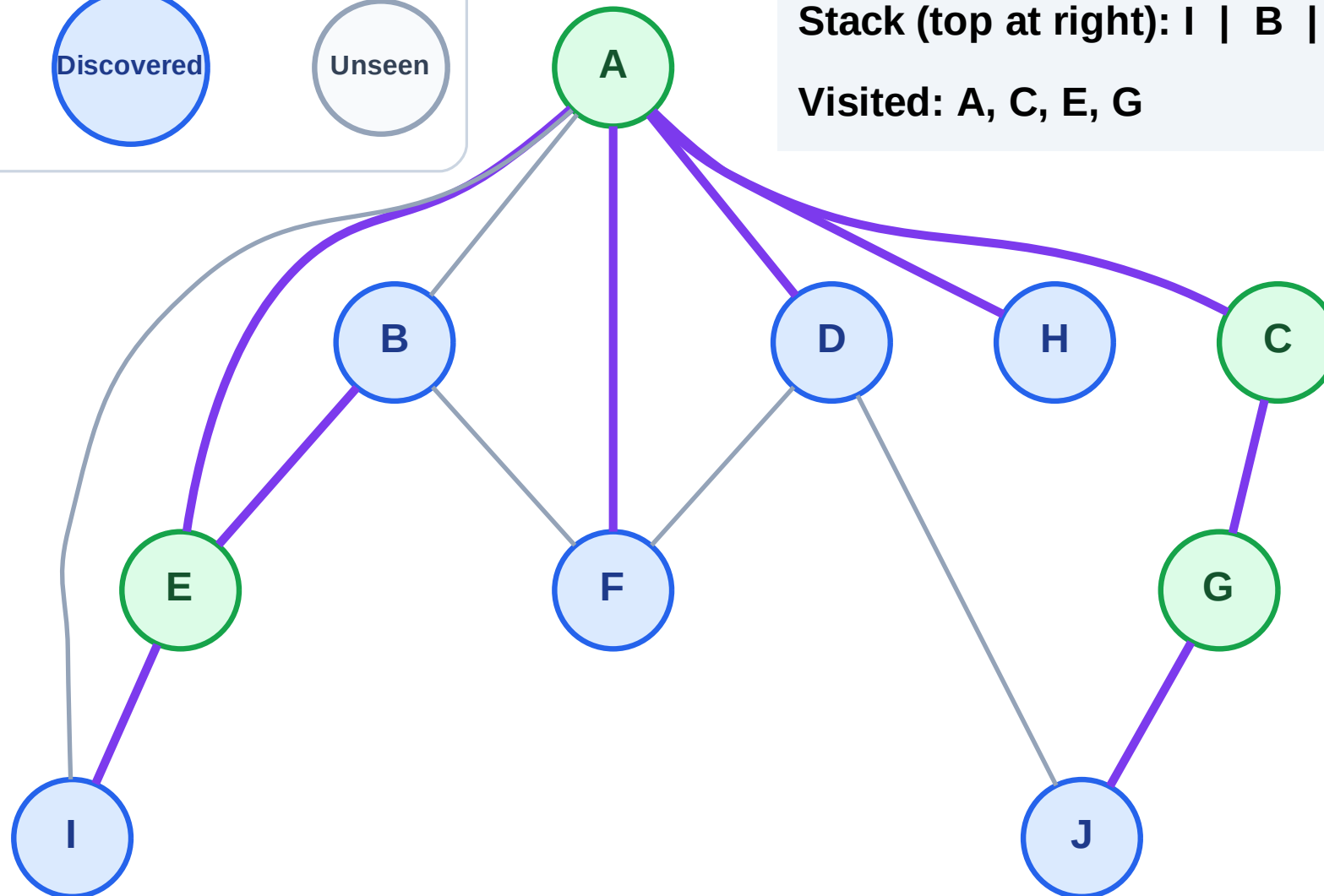
Step 9: Discover J from G

Legend



Stack (top at right): I | B | H | F | D | J

Visited: A, C, E, G



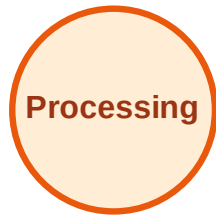
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



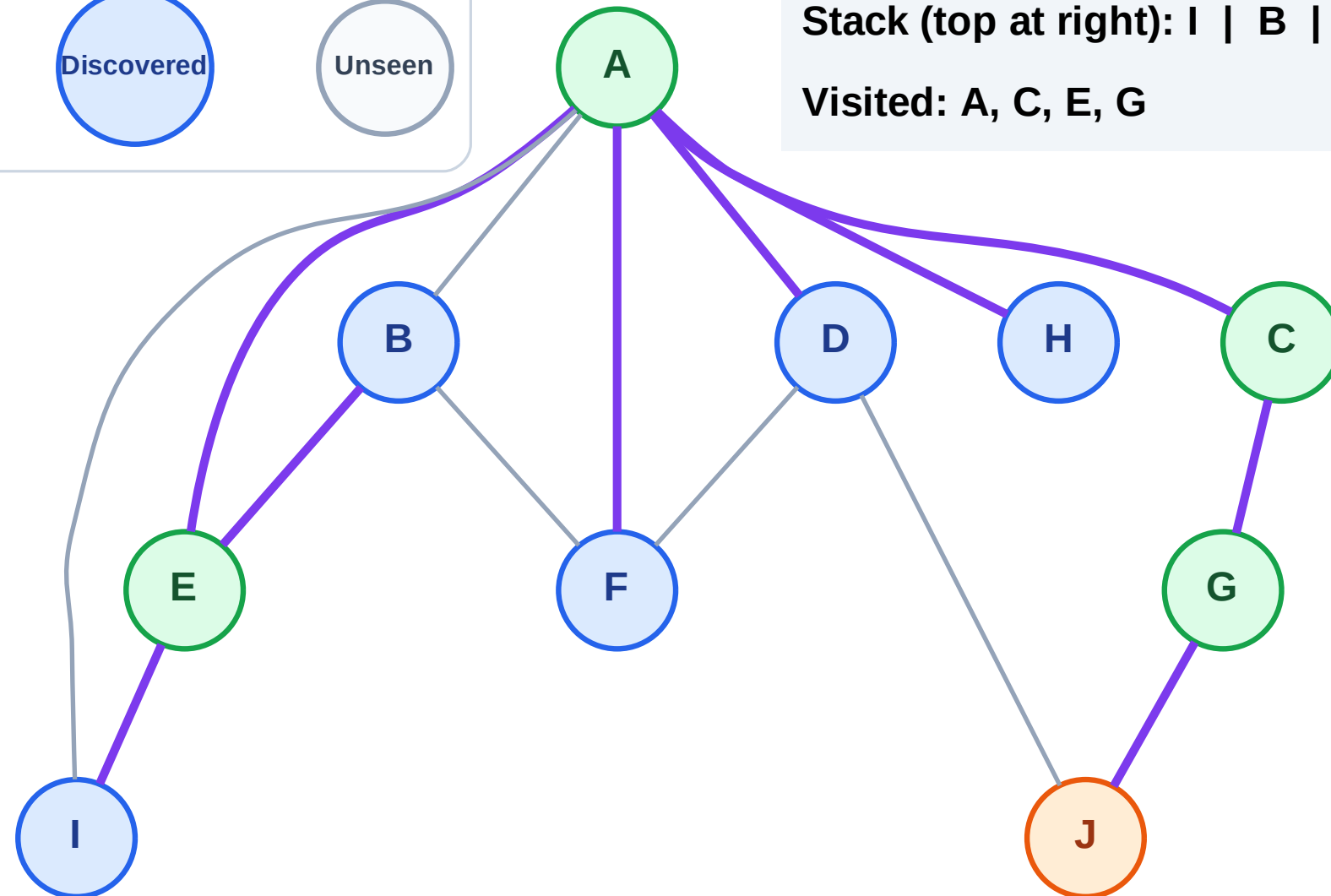
Discovered



Unseen

Stack (top at right): I | B | H | F | D

Visited: A, C, E, G



DFS Traversal

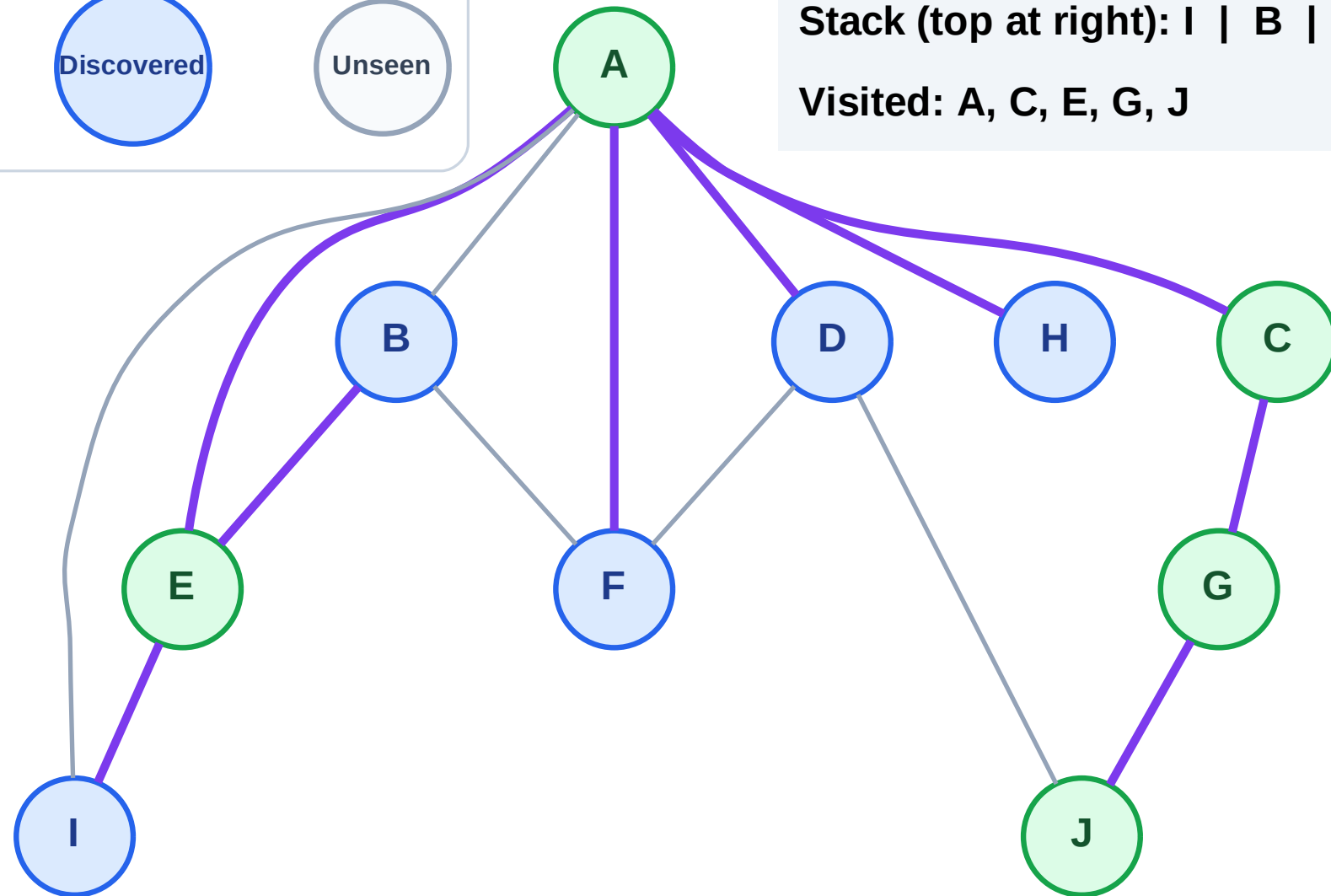
Step 11: No new neighbors from J

Legend



Stack (top at right): I | B | H | F | D

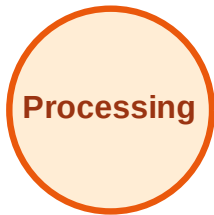
Visited: A, C, E, G, J



DFS Traversal

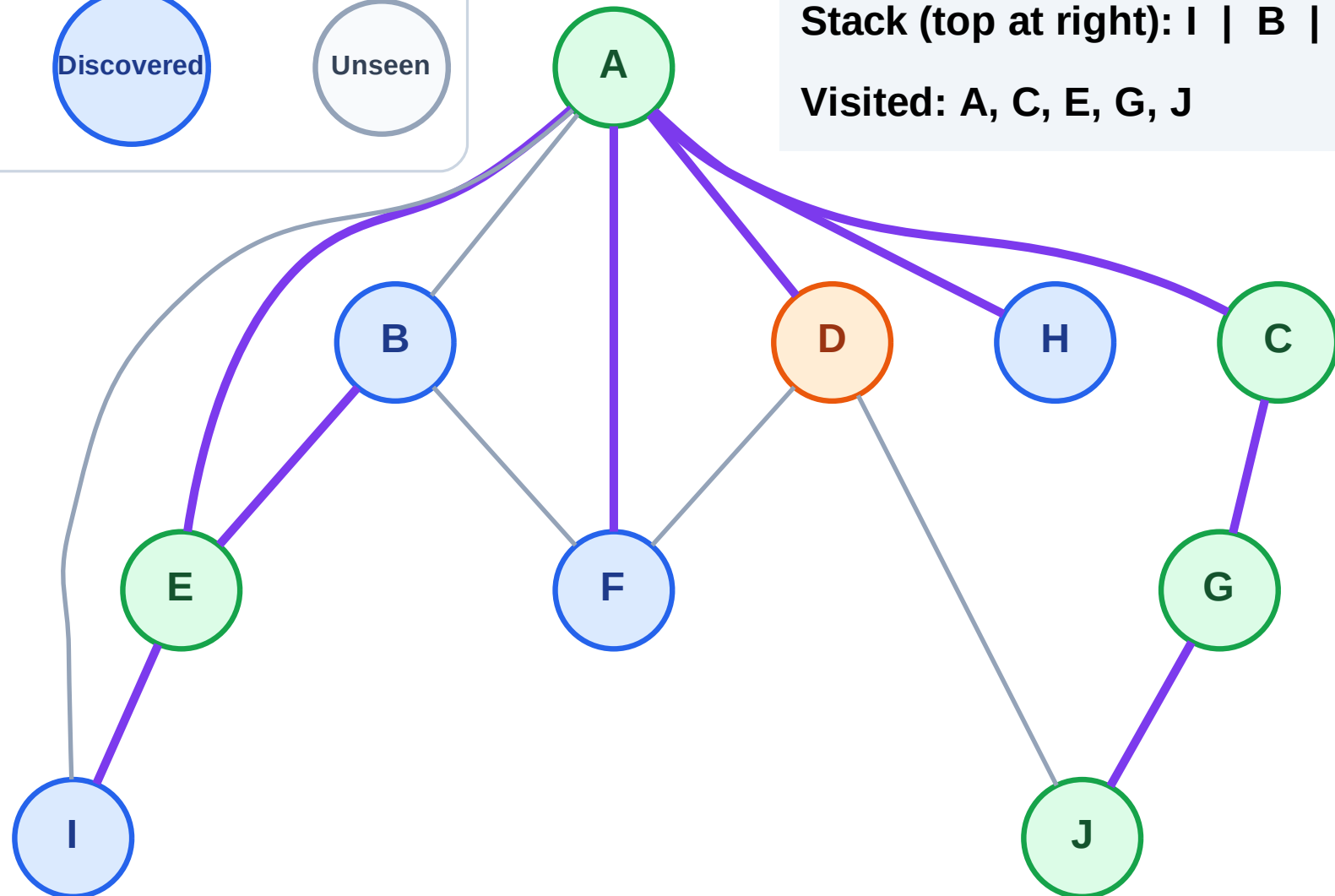
Step 12: Process node D

Legend



Stack (top at right): I | B | H | F

Visited: A, C, E, G, J



DFS Traversal

Step 13: No new neighbors from D

Legend

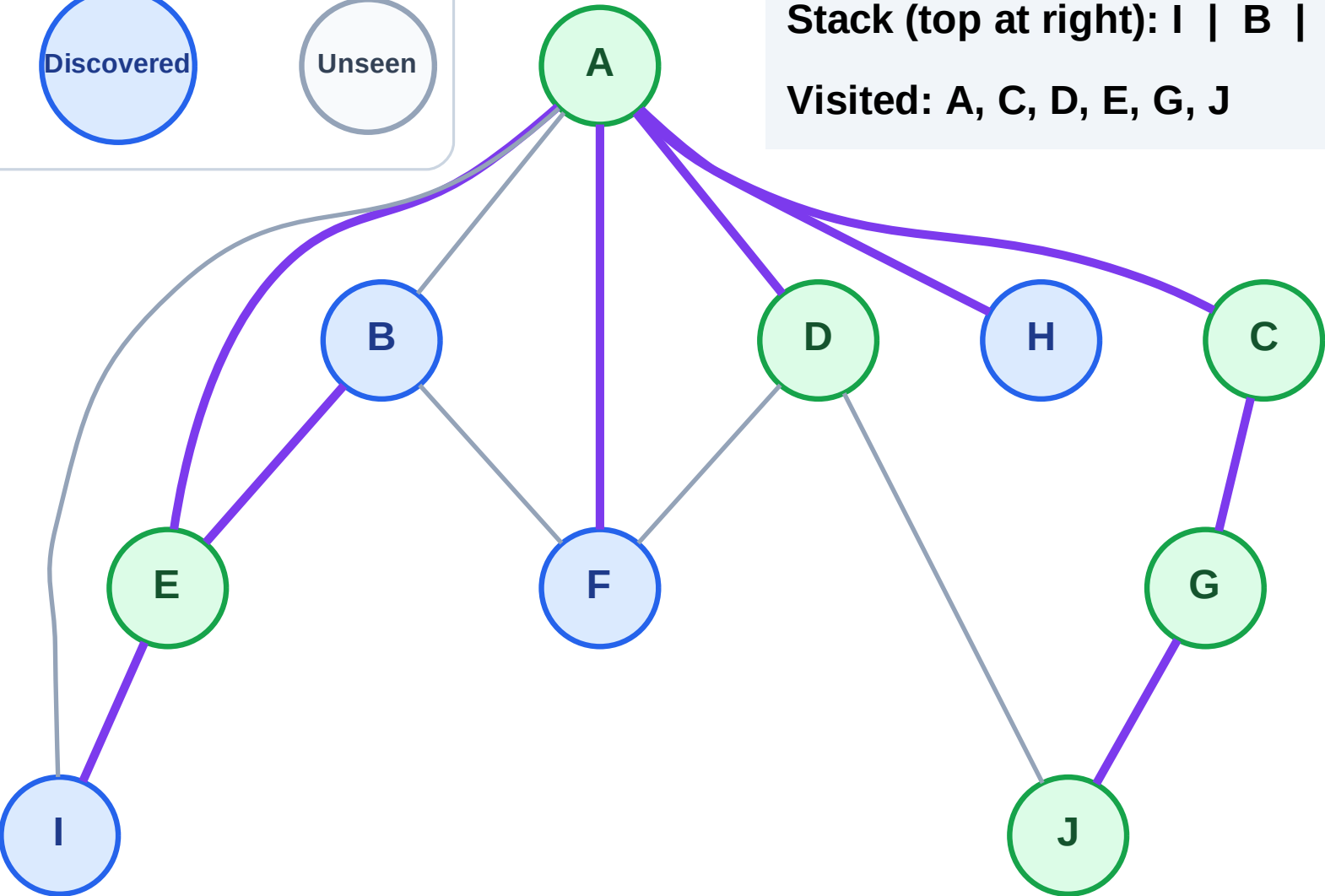
Complete

Processing

Discovered

Unseen

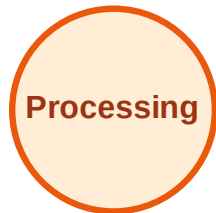
Stack (top at right): I | B | H | F
Visited: A, C, D, E, G, J



DFS Traversal

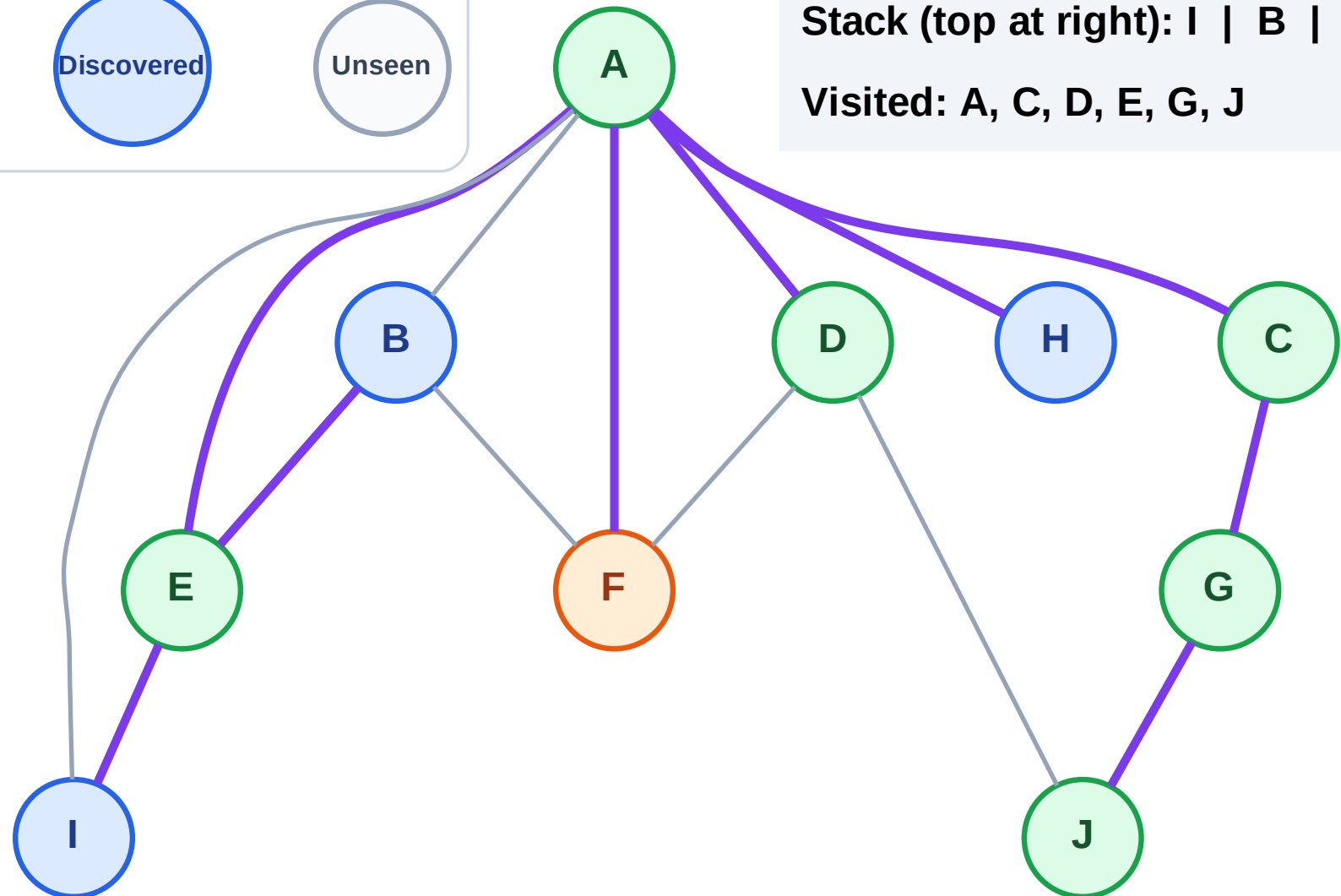
Step 14: Process node F

Legend



Stack (top at right): I | B | H

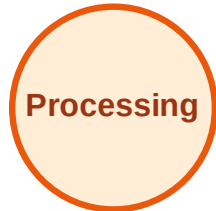
Visited: A, C, D, E, G, J



DFS Traversal

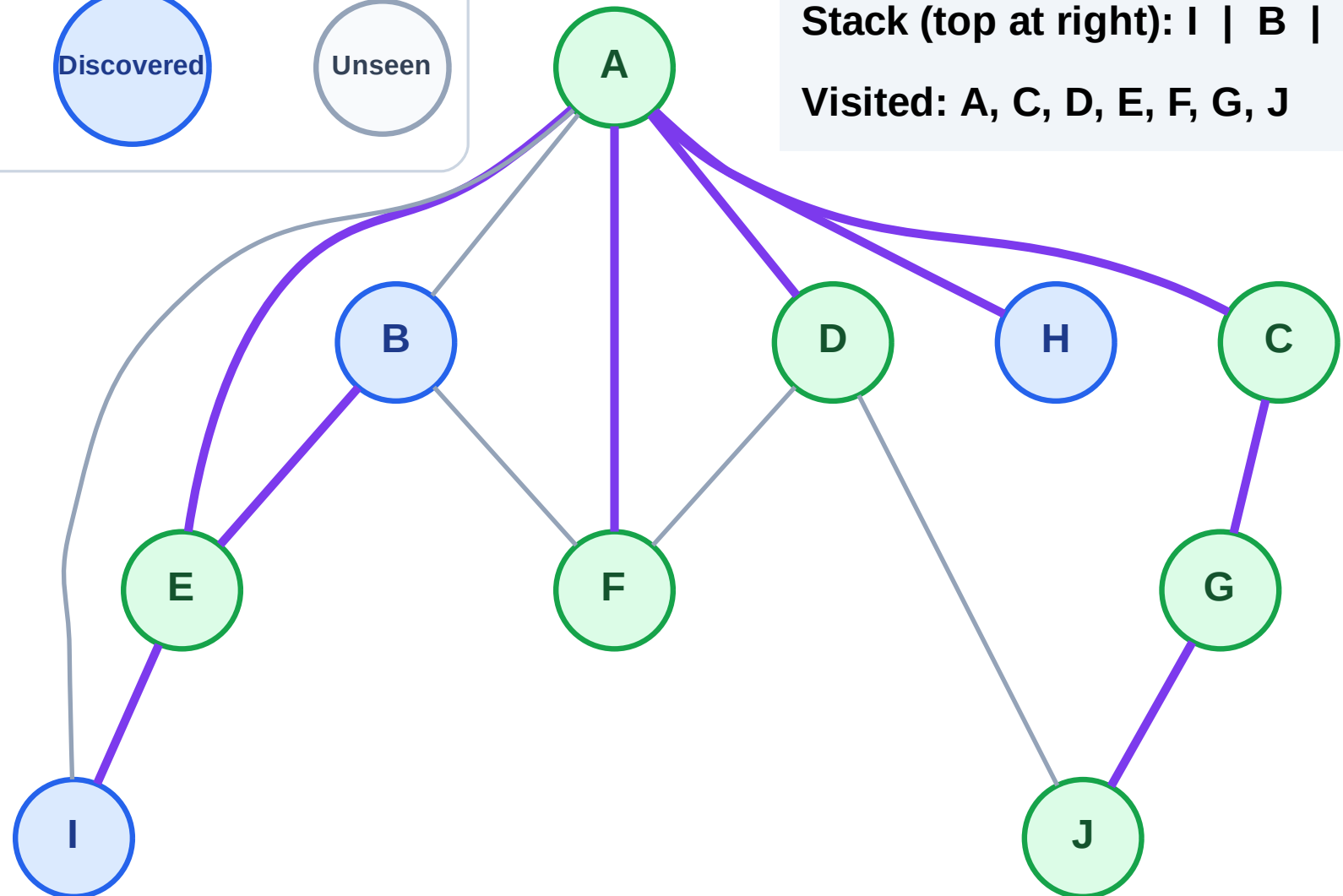
Step 15: No new neighbors from F

Legend



Stack (top at right): I | B | H

Visited: A, C, D, E, F, G, J



DFS Traversal

Step 16: Process node H

Legend

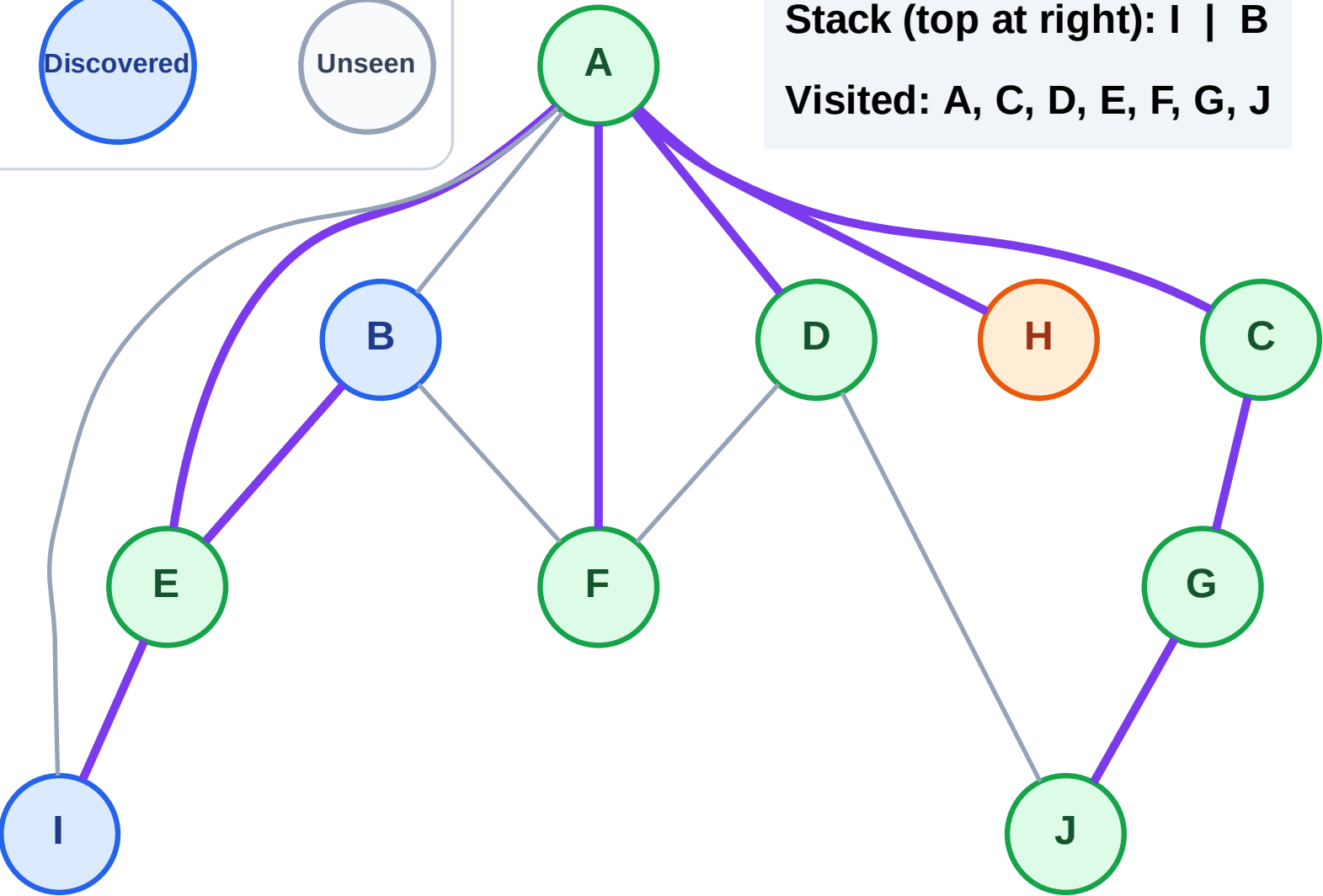
Complete

Processing

Discovered

Unseen

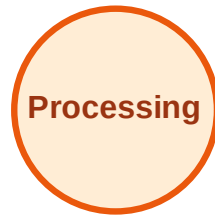
Stack (top at right): I | B
Visited: A, C, D, E, F, G, J



DFS Traversal

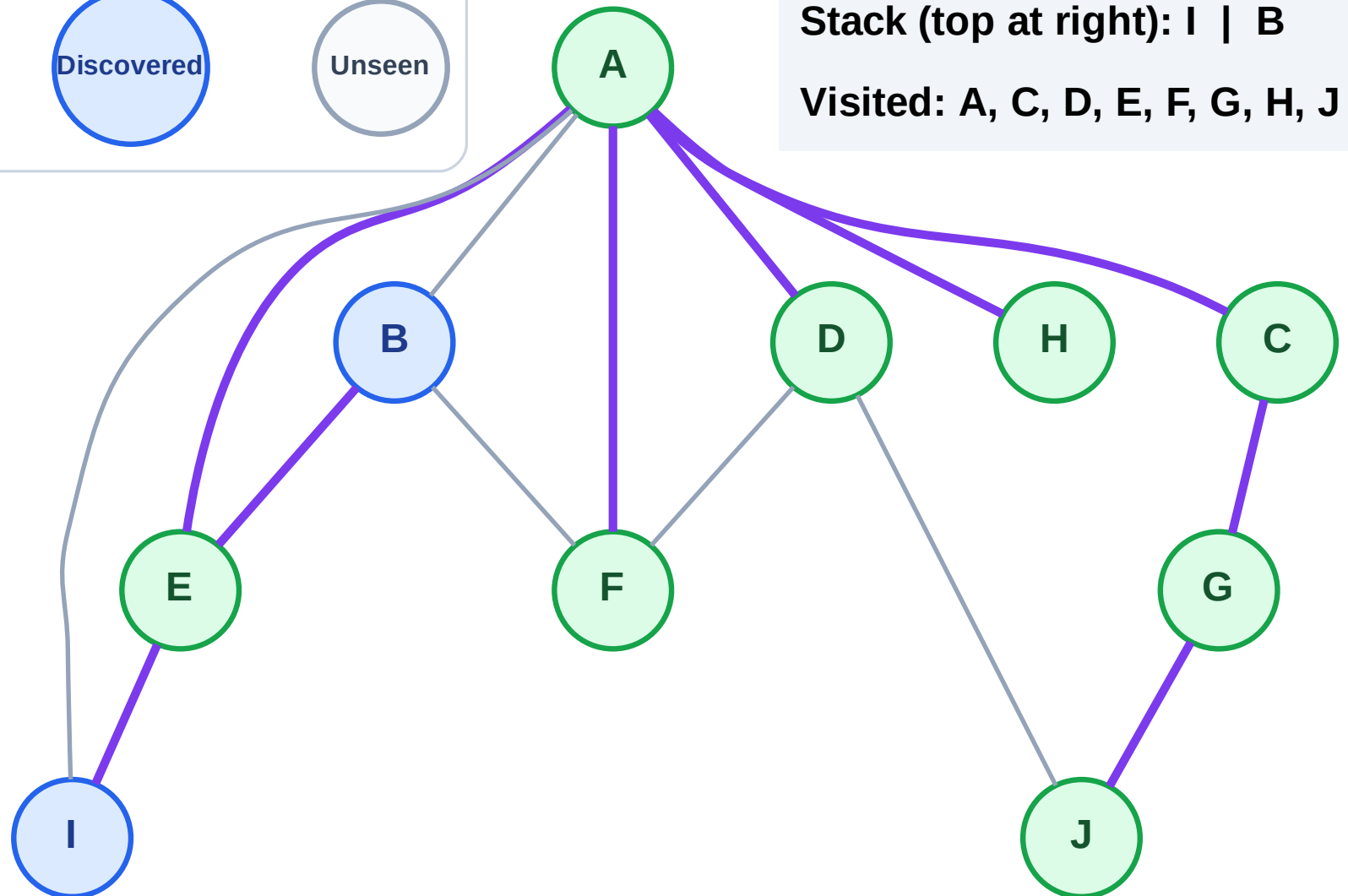
Step 17: No new neighbors from H

Legend



Stack (top at right): I | B

Visited: A, C, D, E, F, G, H, J



DFS Traversal

Step 18: Process node B

Legend

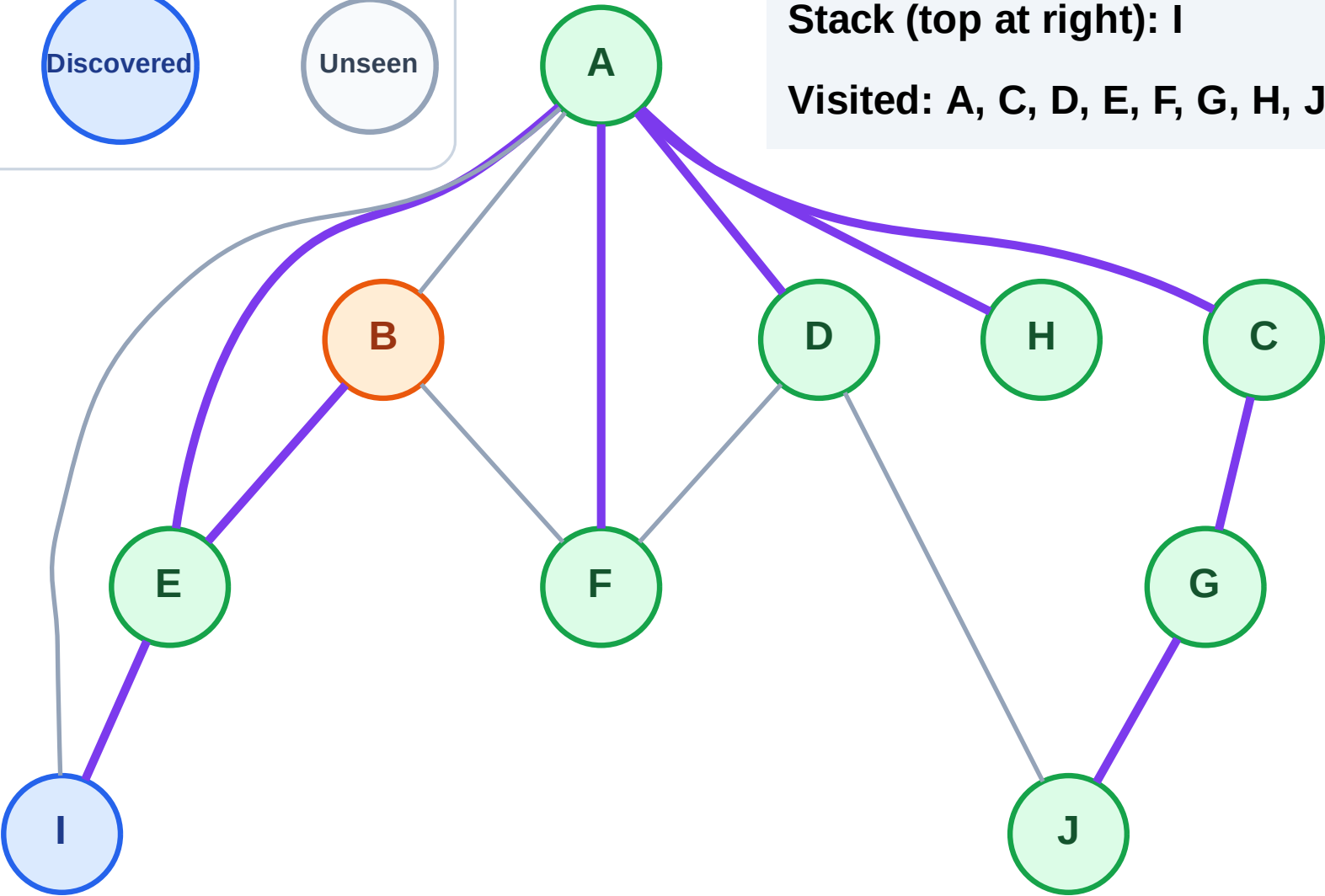
Complete

Processing

Discovered

Unseen

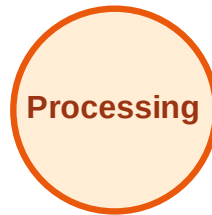
Stack (top at right): I
Visited: A, C, D, E, F, G, H, J



DFS Traversal

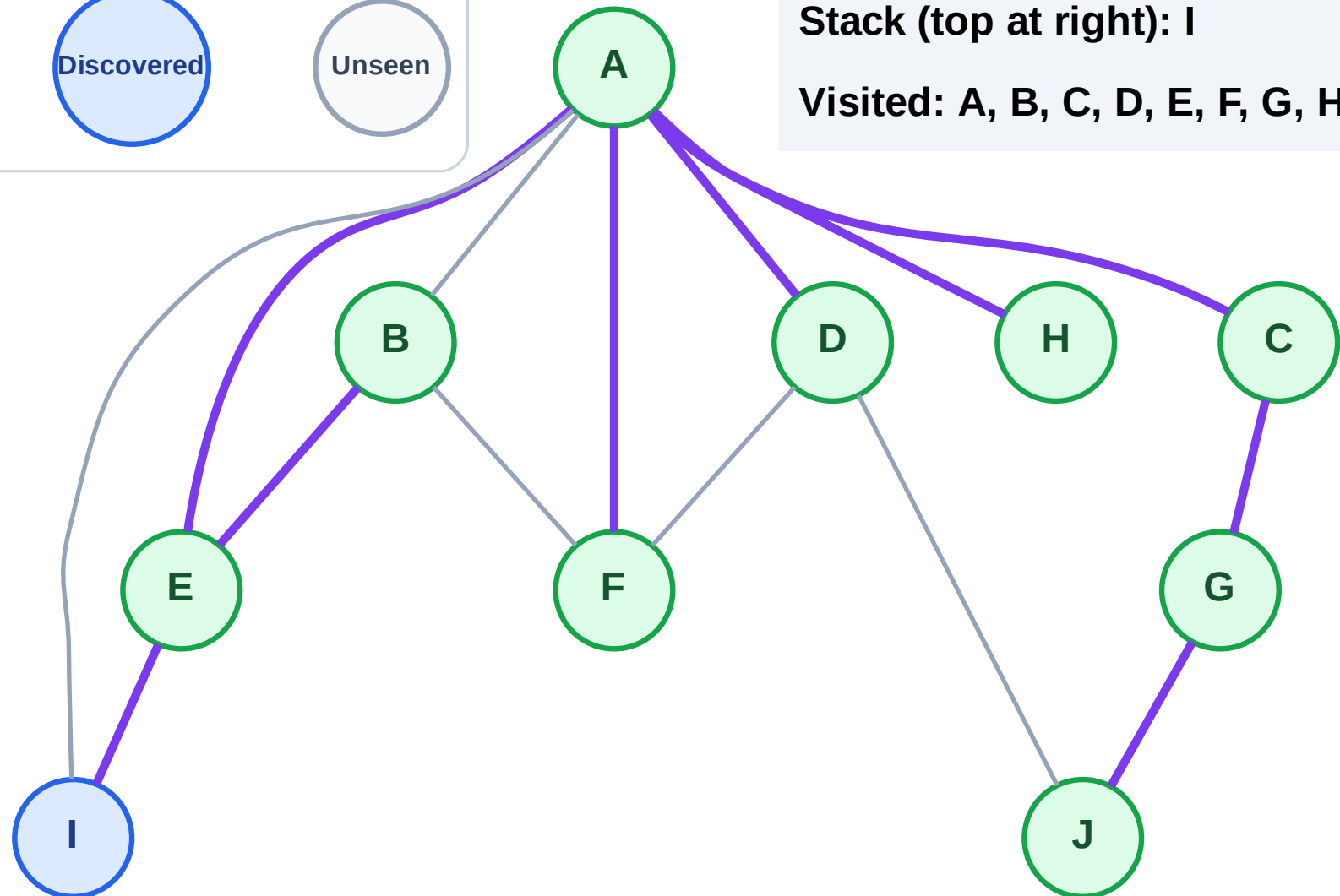
Step 19: No new neighbors from B

Legend



Stack (top at right): I

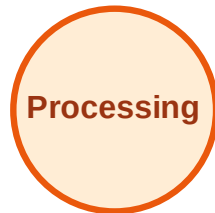
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

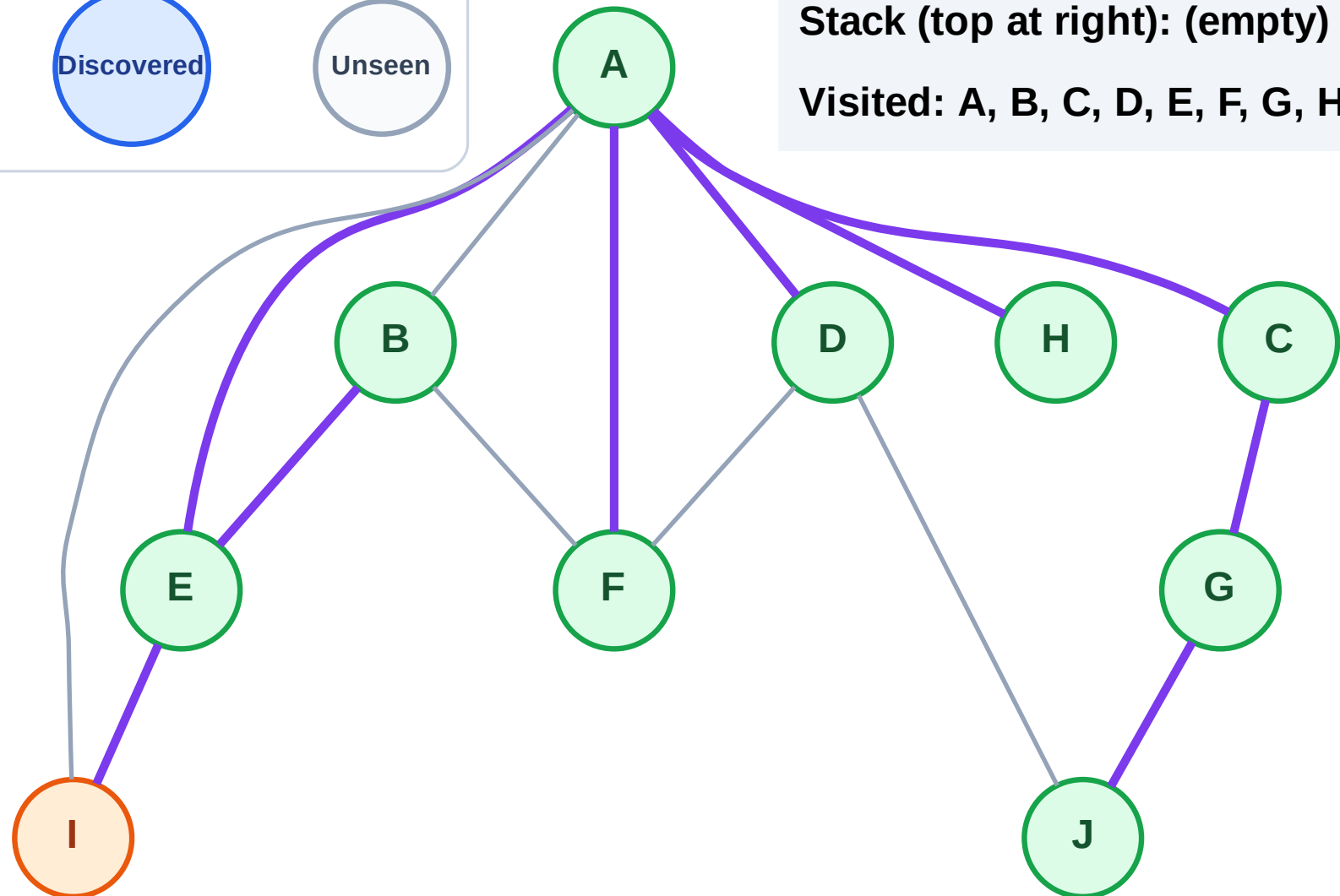
Step 20: Process node I

Legend



Stack (top at right): (empty)

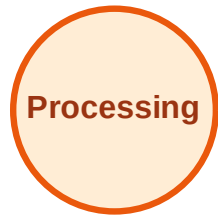
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

